Extreme Heat and Labor Market Dynamics*

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Abstract

This paper investigates the effects of extreme heat on the Brazilian labor market, combining climate data with employer–employee administrative records covering the universe of formal establishments over two decades (2000–2019). We estimate a fixed-effects panel specification for short-run responses and a long-differences approach for medium-run effects. Aggregate responses to heat shocks are null and mask substantial heterogeneity across productivity levels. In the short run, additional days of extreme heat reduce employment and raise the exit probability of the least productive establishments. Plants in the middle and upper parts of the productivity distribution expand their workforce and become less likely to close – possibly due to a greater short-run adaptive capacity – which highlights a redistributive dynamic in local labor markets triggered by high temperatures. In the medium run, however, municipalities exposed to larger decennial increases in extreme heat experience a contraction in their stock of establishments. The contraction operates through entry rather than exit, and is concentrated among high-productivity plants in agriculture and manufacturing. This effect is accompanied by rising employment concentration, driven by the consolidation of high-productivity incumbents. Extreme heat thus reallocates activity toward more productive establishments in the short run, while over longer horizons this reallocation compounds into a more concentrated market structure and discourages the creation of new establishments, a result with potential implications for aggregate productivity growth in emerging economies.

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1 Introduction

Global warming has led to a growing frequency of heat waves. Each year, countries face record-breaking temperatures (The Economist, 2023), with climate scientists projecting increasing exposure to extreme heat events (Zeppetello et al., 2022). In Brazil, where high baseline temperatures already cover a large part of the territory, the number of days of heat waves doubled between the 1990s and the 2010s (Ministry of Science, Technology and Innovation, 2023).[1]

The increase in extreme temperatures can have profound negative effects on the economy.[2] In agriculture, where the effects of temperatures have been widely documented, extreme heat reduces crop yields by accelerating evapotranspiration and disrupting plant growth cycles (Schlenker et al., 2005; Deschênes and Greenstone, 2007; Burke and Emerick, 2016; Blanc and Schlenker, 2017; Pereda and Alves, 2018). At the individual level, heat also affects cognitive and physical performance, especially in outdoor settings or poorly ventilated environments, leading to a reduction in labor productivity (Dell et al., 2014). As a result, high temperatures decrease labor supply and local demand, increase absenteeism, and raise health-related expenditures (Seppänen et al., 2006; Zivin and Neidell, 2014; Somanathan et al., 2021; Liu et al., 2023). Given these broad channels of impact, businesses themselves are likely to be affected by rising temperatures. Previous research has shown that heat can also increase energy expenditure and reduce manufacturing and services output (Dell et al., 2014). Yet, despite these rapid advances, the heterogeneity of these effects across plant characteristics remains underexplored, as do the impacts in developing countries.

This paper addresses these gaps in the literature by investigating the effect of increases in temperature across the full universe of establishments in Brazil, and examining the extent to which extreme heat reshapes labor market dynamics, both in the short and the medium run, and how the effects depend on establishments' productivity level. A large strand of

[1] The Brazilian Meteorological Institute (INMET) defines a heat wave as a period of at least 5°C above the monthly average for at least three consecutive days in a given area of coverage.

[2] High temperatures are also associated with an increase in heat-related illnesses and mortality (Dos Santos et al., 2024; Climate Change Service, Copernicus, 2023). Also, a body of literature linking temperature and economic outcomes reveals strong relationships between extreme heat and energy consumption, crime, political conflicts, and students' performance (Dell et al., 2014; Nguyen, 2023; Melo and Suzuki, 2023).

environmental and labor economics literature has investigated the relation between firms and temperatures in developed countries. These studies have found that heat can increase labor market concentration, disrupt supply chains, reduce aggregate productivity, employment, and earnings (Auffhammer et al., 2017; Goicoechea and Lang, 2023; Ponticelli et al., 2023; Addoum et al., 2023; Acharya et al., 2024; Caggese et al., 2024; Xiao, 2024; Costa et al., 2024; Jin et al., 2025; Cascarano et al., 2025).[3] Importantly, these effects are heterogeneous depending on establishments' features, which may spill over and alter labor market structure: heat shocks can drive employment reallocation toward larger and more productive firms, promote shifts in employment across establishments within multi-establishment firms, and, when establishments adapt, lead to increases in the capital-labor ratio (Ponticelli et al., 2023; Costa et al., 2024; Acharya et al., 2024; Xiao, 2024; Tarsia, 2025).

Evidence for developing countries is less extensive, but understanding how climate change can affect these countries is equally crucial. Low- and middle-income countries (LMICs) typically experience higher baseline temperatures, have less access to adaptive infrastructure, and have a lower-skilled workforce, factors that may amplify establishments' and workers' vulnerability (Bathiany et al., 2018; Acevedo et al., 2020). In addition, firms in these economies often operate with narrower profit margins, limited access to finance, lower technological capabilities, and weaker institutional support, which may further constrain their ability to adapt to environmental stressors (Goicoechea and Lang, 2023; Aguilar-Gomez et al., 2024). Prior research has shown that rising temperatures affect plants' outcomes – including total factor productivity (TFP), employment, delinquency rates, absenteeism levels, and survival rates.[4] These adverse effects tend to be more pronounced among smaller, less productive, private, and more regulated establishments (Zhang et al., 2018; Chen and Yang, 2019; Somanathan et al., 2021; Colmer, 2021; Aguilar-Gomez et al., 2024; Xie, 2024a). This heterogeneous adjustment not only highlights the unequal burden of climate shocks between firms but also provides evidence of a potential redistributive mechanism, where

[3]One exception is the study with public companies from the United States, of Addoum et al. (2021), where the authors find that firms' productivity, sales, and revenues are unresponsive to extreme heat days.

[4]Importantly, the evidence for manufacturing establishments is mixed: while Zhang et al. (2018), Chen and Yang (2019), and Somanathan et al. (2021) find that the overall impacts of higher temperatures on plants are negative, Colmer (2021) states that manufacturing can be positively affected by absorbing local agricultural downturns.

extreme events accelerate the shift of resources toward more resilient and productive firms, with implications for aggregate employment and productivity.

We contribute to this literature by providing new micro-level analysis of the heterogeneous effects of heat in one of the largest middle-income economies in the world. An important contribution of this paper lies in the use of detailed employer-employee data that overcomes many of the limitations faced by previous studies. Much of the existing evidence relies on datasets restricted to specific sectors – such as manufacturing or agriculture – or on firm surveys that impose size restrictions. By leveraging a rich administrative dataset covering the entire universe of formal establishments, we move beyond these sectoral and size constraints to examine how temperature shocks affect the economy as a whole.[5]

To study short-run responses, we employ a fixed-effects model that exploits plausibly exogenous variation in the annual number of days exceeding a microregion-specific temperature threshold, and then interact this variable with a productivity indicator to assess heterogeneous impacts along this dimension. Specifically, we look at how establishment size (measured by the number of employees) and survival rates respond to yearly variations in the number of days with anomalous temperatures (*Heat Days*). We complement these estimates with a long-differences approach at the municipality level to capture medium-run dynamics. This strategy allows us to study whether the yearly effects persist by observing how the differences in averages between 2000-2009 and 2010-2019 in the number of Heat Days affected our labor market outcomes in the same period, after controlling for region-specific common trends.

This paper documents three main findings. First, aggregate null effects on establishment outcomes conceal a sharp gradient across productivity levels. Ten additional *Heat Days* in a year reduce the size of establishments in the bottom decile of the residual wage distribution by 0.70% and raise their annual probability of closure by 0.26 percentage points, a 2.44% increase over their baseline exit rate. Plants in the middle and upper deciles move in the

[5]Regarding evidence from Brazil, to the best of our knowledge, this is the first paper to examine the impacts of high temperatures on establishments. Existing studies linking temperature shocks to labor market outcomes focus exclusively on the individual level: Oliveira et al. (2021) analyze whether elevated temperature affects non-agricultural monthly wages in the short run, and finds that wages decrease as a mechanism for dealing with these shocks; while Xie (2024b), in turn, investigates the effects of extreme heat during the 1990s on the probability of dismissal among formal manufacturing workers, and finds an increase in the likelihood of layoffs, particularly for workers engaged in routine-manual-intensive occupations.

opposite direction on both margins, expanding employment by 0.24% to 0.34% and becoming up to 0.13 percentage points less likely to shut down. These results are consistent with a short-run redistributive mechanism in which climate shocks accelerate the reallocation of resources from less to more productive plants. Additionally, these short-run shocks also increase the share of more educated workers and the probability of importing for high-productivity establishments, a pattern consistent with a differential adaptive capacity by these plants.

Second, the intensity of this response varies systematically across sectors. Agriculture is the most affected on both margins: ten additional *Heat Days* contract bottom-decile plants by 1.22% and raise their exit propensity by 0.55 p.p., or 3.93% of the baseline rate, with adverse effects reaching up to the fourth decile and gains at the top too small to offset them. Manufacturing follows, combining a monotonic size gradient that runs from -0.98% to $+0.89\%$ with a 0.30 p.p. rise in closure risk at the bottom. Establishments in the services sector, which accounts for the majority of the sample, show the smallest coefficients for both losers and winners, and their results are similar to the pooled ones.

Third, the medium run reverses the aggregate picture: the effect of prolonged exposure is itself negative rather than redistributive. A ten-day increase in the decennial average number of *Heat Days* reduces the stock of active plants at the municipality level by 2.10%. This contraction does not come from more establishments shutting down, as exit coefficients are small and statistically insignificant, but from suppressed creation, as the number of new establishments entering falls by 3.42%. Disaggregating by productivity shows that the entry response strengthens as one moves up the distribution, with declines of 3.20% and 4.12% in the fourth and fifth quintiles, and the losses in the establishment stock are concentrated at the top (2.64% in the fifth quintile). These patterns are more pronounced in agriculture and manufacturing, where entry at the top quintile falls by 8.30% and 4.51%, respectively. This weaker entry of high-productivity plants is accompanied by rising employment concentration within local industries, driven by the consolidation of high-productivity incumbents rather than by any expansion of high-productivity establishments as a group.

Our work is connected to the literature on the effects of labor market reallocation driven by temperature shocks. Studies in developed economies have found that excess heat can

trigger a reallocation of employment from more to less affected areas, from single- to multi-location firms within the same affected area, and from small to large plants (Ponticelli et al., 2023; Acharya et al., 2024; Costa et al., 2024). In addition, part of the literature has demonstrated that heterogeneous impacts across industries, seasons, and firm sizes can generate redistributive effects within local economies (Colmer, 2021; Addoum et al., 2023; Caggese et al., 2024; Albert et al., 2024; Jin et al., 2025).[6]

Building on this literature, a key contribution of our analysis is to examine not only heterogeneities across sectors but also differences between low- and high-productivity establishments, thereby shedding light on how abnormal temperature shocks can trigger redistributive dynamics within the economy. A few papers have explored how heat affects establishments with different levels of productivity. Xie (2024a) shows that, in the universe of medium and large manufacturing firms in Indonesia, resources are reallocated from less productive to more productive firms. Also, Tarsia (2025) finds that in several European countries, firms in lower total factor productivity deciles are negatively affected by rising temperatures, while those in the upper deciles improve their performance – likely reflecting reallocation patterns. Beyond documenting similar results regarding how losses among low-productivity plants are offset by gains among high-productivity ones, we go further by examining productivity heterogeneity within sectors, highlighting how the relationship between productivity and heat shocks differs across industries. Also, we provide novel evidence that these employment movements across the productivity distribution are not homogeneous, as those at the top rearrange their workforce composition in the short run to more skilled labor. Moreover, most existing studies use the *firm* as the unit of observation – mainly due to data restrictions – which could bias the results, since a single firm can hold multiple smaller subsidiaries that may operate almost independently. By focusing on the *establishment* level – the most granular unit of a business – we are able to overcome these potential issues and yield more precise estimations.

We also add to a smaller strand of the literature that goes beyond the impact of annual temperature fluctuations and observes how medium- or long-run changes in temperatures

[6]The paper from Albert et al. (2024) is a particular case; although it looks at the impacts of droughts, instead of temperature, it shows that excess dryness in Brazil causes labor reallocation away from agriculture and services to manufacturing.

alter the labor market dynamics, approaching the notion of *climate change* effects, rather than annual weather variations. In this sense, studies at the macro level have demonstrated that the medium-run effects of temperature might decrease per capita income and investments (Dell et al., 2012; Acevedo et al., 2020). More granular evidence remains mixed, depending on the time span, climatic phenomena, and country context. Liu et al. (2023) find that long-run temperature variations in India reallocate employment toward agriculture. Although focusing on dryness instead of temperature, but particularly relevant to our analysis for examining Brazil, Albert et al. (2024) shows that excess dryness over a decade reduces agricultural employment in the most affected areas. In the US, Ponticelli et al. (2023) document that long-run temperature increases expand employment in large manufacturing plants. In Italy, Cascarano et al. (2025) show that persistent exposure to hot temperatures reduces firm entry and raises exit rates, with effects concentrated in warmer Mediterranean areas and among micro firms. Leveraging the extended time span of our data, we add to this literature by providing new evidence on the medium-run effects of temperature change in a large middle-income country – an underexplored context – showing that suppressed entry among high-productivity plants coincides with rising local employment concentration, a novel structural channel through which heat exposure can compound over time. Our findings, therefore, expand the empirical understanding of how climate change reshapes labor market dynamics in emerging economies.

The rest of this paper is organized as follows. Section 2 describes the climate and establishment-level datasets, presents descriptive statistics, and explains our measure of establishment productivity. Section 3 outlines the empirical strategies, including the annual panel specification for short-run effects and the long-differences approach for medium-run effects. Section 4 presents the short-run results, and Section 5 the medium-run results. Finally, Section 6 concludes.

2 Data and Descriptive Statistics

2.1 Climate Data and Descriptive Statistics

We use climate data provided by ERA5-Land. This dataset enables us to observe hourly climate variables with $0.1° \times 0.1°$ horizontal resolution (that is, grids of approximately 11 km $\times$ 11 km), and has the advantage of using reanalysis, a methodology that optimally combines observations and models to provide consistent and high-resolution estimates. We extract temperature and precipitation data for each Brazilian grid cell and spatially interpolate them to municipalities, thereby constructing a municipality-level panel dataset of climate variables covering the two decades (2000–2019) in which heat waves became increasingly frequent in Brazil (Marx et al., 2021).[7]

Brazil's vast territory is characterized by significant climate heterogeneity, with especially high baseline temperatures in the central and northern regions of the country. This heterogeneity motivates defining extreme heat relative to local rather than national conditions, and the Brazilian administrative geography provides a natural unit for doing so. The country is divided into five regions and 27 states.[8] Each state is further partitioned into mesoregions, which are in turn subdivided into microregions – groups of municipalities sharing similar natural endowments and productive structures. Each state has, on average, 21 microregions, totaling 558 microregions, each comprising roughly ten of Brazil's 5,570 municipalities.[9]

Our main climate variable is called *Heat Day*. In order to define a Heat Day, and to better capture the climate diversity of Brazil, given its large territorial dimensions, we first obtain the daily distribution of the maximum temperature for all municipalities in each microregion of Brazil during the period 2000-2019.[10] This process generates 558 maximum-temperature

[7]Temperature data represent the temperature of air at 2m above the surface of land, sea, or inland waters, and are calculated by interpolating between the lowest model level and the Earth's surface, taking account of the atmospheric conditions. Precipitation represents the accumulated liquid and frozen water, including rain and snow, that falls to the Earth's surface.

[8]The five regions are North, Northeast, Mid-West, Southeast, and South.

[9]Brazilian microregions are closest to commuting zones in the US, and it groups together economically integrated contiguous municipalities with similar geographic and productive characteristics (Martinez and Mello, 2024).

[10]Brazil is the fifth largest country in the world, in square kilometers, and the second in length extension (that is, the distance between the southernmost point and the northernmost point).

distributions, one for each microregion. A municipality is then considered exposed to a Heat Day if its maximum temperature on a given day exceeds the 95th percentile of the distribution of the microregion to which it belongs. These microregion-specific thresholds ensure that extreme heat is defined relative to local climatic conditions rather than to a national benchmark, so that a Heat Day carries a comparable meaning across hot and mild areas. The maps below show the 95th percentile thresholds by microregion and the yearly average number of *Heat Days* by municipality.

Figure 1: Thresholds (°C) by Microregions

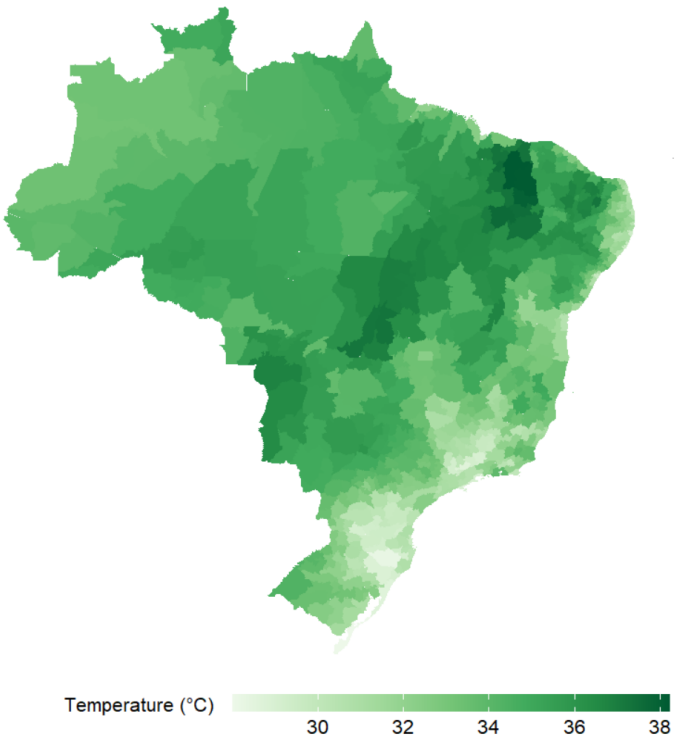


Figure 2: Annual Average Number of *Heat Days* by Municipality

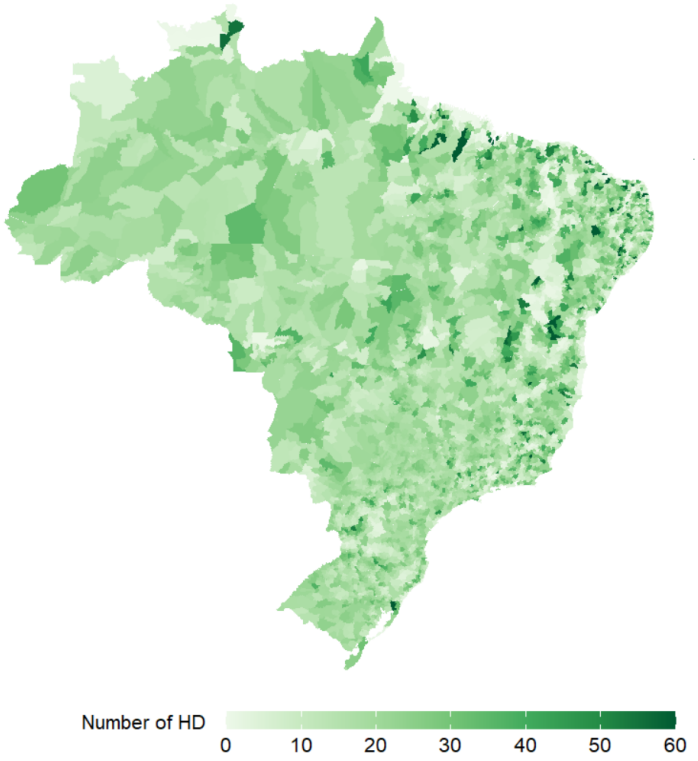


Figure 1 shows that the microregion-level thresholds are highest in the North, the interior of the Northeast, and in central Brazil, reaching up to 38°C in some microregions, and lowest in the South and along the Southeast coast. However, Figure 2 exhibits no macroregional gradient: structurally hot and structurally mild areas record broadly similar counts of Heat Days. This is by construction and is the intended property of the measure – because thresholds are set within microregions over the full sample period, baseline climate is differenced out of the exposure variable. Figure A.1 and Figure A.2 document how exposure evolved over the period: the first maps the decennial growth in *Heat Days* across municipalities, which was positive in most of the country though uneven across space, while the second plots annual regional averages, showing an upward trend in all five regions alongside substantial year-to-year volatility.

2.2 Establishment Data

To estimate the impact of extreme heat on establishments' performance in Brazil, we utilize employer-employee administrative data from RAIS (*Annual Social Information Report*). RAIS is a comprehensive administrative database managed by the Brazilian Ministry of Labor and Employment, capturing detailed information on formal labor market activities across the country. Establishments operating in Brazil are legally required to submit accurate and complete RAIS reports annually, which include data on all employees on their payroll during the previous year. Each line in the database is an employer-employee match (a spell), including a unique worker ID and a unique establishment ID, allowing us to trace establishments over time.[11]

For each spell, the dataset includes detailed worker information (e.g., age, gender, race, and educational level), establishment-related variables (e.g., specific industry, municipality, and legal nature), and job-specific information (e.g., tenure, salary, detailed occupation, start and end dates). By aggregating worker-level information, we construct establishment-level characteristics such as size, average wages, and workforce composition in terms of occupation and educational level. We exclude financial and public firms, as well as establishments with an average wage of zero. Finally, to incorporate weather data, we match establishments' municipalities to our municipality-level climate dataset. This process generates an establishment-level panel covering the years 2000 to 2019, enriched with weather variables.

2.2.1 Productivity Measure

The RAIS dataset does not provide establishment-level output information (e.g., revenues or sales), which makes it challenging to measure productivity. However, a large strand of theoretical and empirical labor literature states that firms with higher average wages tend to be more productive.[12] To infer establishment productivity over time, we create a proxy based on establishments' residual average wages within local labor markets. We recover

[11] An important limitation of RAIS is that it does not cover the informal sector, which corresponded to around 30% to 40% of the Brazilian workforce during our analysis period.

[12] The main reasons are that they employ a larger share of high-wage – and more productive – workers, but also because these firms pay a higher wage premium, and these workers sort into these firms (Abowd et al., 1999; Criscuolo et al., 2021; Berlingieri et al., 2024).

these residual wages by running the following regression:

$$\ln(wage_{it}) = \boldsymbol{X}'_{it}\boldsymbol{\beta}_t + \alpha_t + \mu_{it} \tag{1}$$

and obtain $\hat{\mu}_{it}$. Specifically, we regress the log wage of establishment i in year t on a vector of establishment-level features of its workforce ($\boldsymbol{X}_{it}$) and year fixed effects.[13] We also allow the parameters $\boldsymbol{\beta}_t$ to vary across years by fully interacting the covariates with year dummies. The residual $\hat{\mu}_{it}$ therefore captures the portion of an establishment's average wage left unexplained by the returns to workforce characteristics prevailing in that specific year (establishment wage premia), which keeps residuals comparable in magnitude across the whole period. We then rank establishments according to their position in the residual wage distribution of their local labor market (LLM), defined as a pair of industry and microregion.[14] Since each establishment operates in a single LLM, this distribution is computed over all of its establishment observations and split into deciles: plants in the top deciles of their LLM are the most productive, and those in the bottom deciles the least productive.

A more precise approach to measuring establishment productivity would be to recover establishment fixed effects (also known as *firm effect*) from AKM regressions (Abowd et al., 1999), isolating the wage premia from worker composition (Kline, 2025). However, AKM identification requirements generate substantial sample losses concentrated among small establishments, which are central to our analysis of heterogeneous impacts. Hence, we rely on residual average wages within microregion-industry cells as our primary productivity proxy, which allows us to retain the full universe of formal establishments. In Appendix C, we detail the AKM estimation procedure, document the characteristics of excluded establishments, and show that our residual wage-based proxy exhibits a strong positive correlation with estimated firm effects, supporting its validity as a productivity measure. Nevertheless, our productivity measure should be interpreted with caution, as our residual average wages reflect not only the firm's productivity but also sorting.

[13] The vector $\boldsymbol{X}_{it}$ contains the shares of high-school graduate workers, high-skill (occupation) workers, and male workers, as well as average tenure, age, and age squared.

[14] We use the definition of CNAE section, which encompasses 21 industries. Since Brazil has 558 microregions, this yields 11,718 potential cells, of which 9,333 remain after excluding those with fewer than ten observations.

Previous literature has shown that heat is negatively associated with firm productivity and average wages (Oliveira et al., 2021; Caggese et al., 2024). This raises a post-treatment concern in our setting: since deciles are assigned from residual wages computed in the same year in which heat is measured, an establishment's productivity rank may itself respond to the treatment.[15] Ranking within industry × microregion cells offers partial protection, as shocks common to all establishments in a cell leave relative positions unchanged, but does not rule out differential effects across establishments of the same cell, which are precisely the ones that would alter the ranking. In line with existing literature, Table B.1 confirms a reduction in firm productivity, yet this does not translate into movement across productivity ranks, as the effects on the cardinal distance between an establishment's deciles in year t and $t+1$, and the probabilities between the same periods of changing, moving up, and moving down deciles are all null. Panel B reveals that *Heat Days* makes establishments in the bottom deciles more likely to move up and less likely to move down, while the reverse holds at the top. Movement is therefore strongest at the tails, where transitions are mechanically restricted to a single direction; across the middle of the distribution, the probability of changing decile falls, indicating greater rank stability. Hence, if anything, heat-induced rank mobility works against, rather than in favor of, a post-treatment interpretation.

2.2.2 Establishment Descriptive Statistics

Table 1 presents descriptive statistics of our sample, highlighting a highly right-skewed distribution in establishment size. The average establishment employs 11.8 workers, while the median is only 3, indicating that most establishments are relatively small. We divide our economy into three main sectors: agriculture, manufacturing, and services. Establishments in the manufacturing sector tend to be larger and pay higher wages, on average. The agriculture sector reports the lowest average wages, followed by services. Our proxy for establishment productivity reveals a pattern consistent with findings in the labor economics literature, which typically document a high correlation between plant size and productivity: smaller establishments tend to be less productive, whereas larger ones exhibit higher pro-

[15] If heat displaced establishments across deciles, we would be conditioning on a post-treatment variable, and the estimated productivity gradient would partly reflect heat-induced reshuffling rather than pre-existing heterogeneity.

ductivity levels. While services account for over 80% of establishments, this sector employs only 66.08% of the workforce and features a relatively more qualified composition (Table A.1).

Table 1: Employment and Wages by Sector and Establishment Productivity Level

Group	Employment			Wages (Monthly BRL)			Total Units
	Mean	SD	Median	Mean	SD	Median	
All plants	11.84	79.94	3	1536.90	1244.18	1290.60	44,391,115
Sector							
Agriculture	13.23	79.74	2	1377.82	1174.78	1135.44	881,385
Manufacturing	25.20	139.19	5	1744.31	1591.85	1402.33	6,610,781
Services	9.41	63.46	3	1503.54	1168.72	1277.76	36,898,949
Productivity							
1st decile	4.43	29.86	2	842.09	283.22	826.00	4,438,018
5th decile	9.21	62.23	3	1282.06	372.56	1233.37	4,432,335
10th decile	27.45	145.23	5	3359.59	2993.59	2673.14	4,429,566

Notes: Economic sectors are classified according to the two-digit CNAE (National Classification of Economic Activities) codes. We classify agriculture as CNAEs 1-3, manufacturing as CNAEs 5-42, and services as CNAEs 43-99. Productivity levels are classified according to deciles of the residual average wage distribution within its local labor market. Wages are in 2019 prices.

2.2.3 Complementary Datasets

We complement our data with two additional datasets. First, we use one wave (2000) of the Brazilian population census (*Censo*), which provides municipality-level characteristics such as population, per capita income, urbanization, educational attainment, and workforce formalization. We use these data to construct the baseline controls for our medium-run estimations, as detailed in Section 3.2. Second, we use an establishment-year panel from the Brazilian Secretariat of Foreign Trade (*SECEX*), which covers every establishment that exported or imported in a given year. These data record only the extensive margin, that is, whether an establishment engaged in international trade in a given year, but not which products it traded or in what quantities or values. We use them to study adjustments in

trade participation in Section 4.2.

3 Empirical Strategies

Our goal is to quantify how an additional Heat Day affects both establishment-level performance outcomes – capturing short-term, transitory effects – and municipality-level aggregated outcomes, which reflect medium-run impacts. To do so, we first estimate establishment-level specifications to assess the effects of heat shocks on employment and the likelihood of closure. Then, we estimate municipality-level specifications to evaluate the decennial consequences of climate change.

3.1 Short-Run Analysis: Establishment-Level Panel

Our baseline specification examines the heterogeneous impacts across productivity levels of an additional day of extreme heat on establishments' intensive and extensive margins (respectively, size and survival), based on the following fixed effects model:

$$Y_{it} = \sum_{d=1}^{10} \beta_d Prod_{it}^{d} \times HeatDays_{mt} + \delta Precip_{mt} + \theta_i + \phi_{jt} + \gamma_{rt} + \varepsilon_{i(m)t} \tag{2}$$

where Y_{it} represents the dependent variable, which corresponds either to the establishment's size, measured by the log number of employees reported at the end of each year t, or to an indicator if the establishment left the panel (dummy that takes the value of 1 if the establishment was no longer in RAIS in year $t+1$). The treatment variable is the number of days per year in which a municipality m was exposed to a Heat Day, represented by $HeatDays_{mt}$. Furthermore, we include a control variable for rainfall ($Precip$), measured as the average cumulative precipitation, to isolate the potential effects of extreme temperatures from those related to variations in rainfall.

Larger and more productive establishments may be better equipped to cope with extreme heat due to greater financial flexibility, more developed infrastructure, and better management practices, while smaller or less productive establishments, which often operate with tighter financial constraints or lower technological adoption and weaker organizational

capacity, would be more vulnerable (Ponticelli et al., 2023; Costa et al., 2024; Xie, 2024a; Bettarelli et al., 2025). To assess productivity-based heterogeneity, we interact the productivity category dummies ($Prod$) with our treatment variable. This interaction allows us to isolate the differential effects of heat exposure across productivity strata. We decompose the interacted coefficients so that each β_d captures the marginal effect of extreme heat over establishments in each of the deciles.

Our key identification assumption is that, conditional on the set of fixed effects described below, variation in the number of *Heat Days* is uncorrelated with any other determinant of establishment outcomes. First, we include establishment fixed effects, θ_i, which absorb all time-invariant establishment characteristics. Since each establishment operates in a single municipality, these fixed effects also absorb time-invariant municipal features – such as geography, climate, infrastructure, and local institutions – that could generate systematically different performance in warmer areas. Second, industry-year fixed effects (ϕ_{jt}) account for sectoral shocks common to all establishments in a given industry, such as changes in demand, input prices, or industry-specific policies.[16] Third, macroregion-year fixed effects (γ_{rt}) absorb annual shocks common to broad areas of the country, including regional business cycles and large-scale climatic phenomena that simultaneously affect entire portions of the territory.[17] The identifying variation is therefore within establishment: we compare the outcomes of a given plant across years in which its municipality was exposed to more or fewer *Heat Days* than usual, relative to establishments in the same industry and macroregion in the same year. Standard errors are clustered at the municipality level.

One potential threat is the evolution of urbanization levels. Urbanization may raise local surface temperatures through the formation of urban heat islands, while also being associated with greater economic activity (Ullah et al., 2023). Rising temperatures could then be endogenous to economic outcomes through unobserved urban development, a form of reverse causality. Three arguments mitigate this concern. First, establishment fixed effects absorb time-invariant urban traits, which matters because urbanization levels change slowly.

[16] The industry-year fixed effects use 3-digit CNAE codes, which account for 285 industries.

[17] Temperature is strongly spatially autocorrelated across neighboring municipalities, so finer geographic-by-year fixed effects would absorb much of the variation in heat exposure that identifies our estimates. We therefore chose to interact year dummies with macroregions.

Second, as argued by Caggese et al. (2024) and Costa et al. (2024), the hypothesis is difficult to reconcile with our findings and with the existing literature, since anomalous temperatures are not generally associated with expansions in establishment output or in local economic activity. Finally, in Table B.2 we test if the lagged realized growth of employment and establishments' stock predicts the variation in *Heat Days* by performing a regression on a municipal-level panel with our treatment as the outcome.[18] If economic activity were driving heat exposure, past economic growth should be associated with subsequent heat. Across all specifications, the estimated coefficients are insignificant, supporting the view that reverse causality is not an issue in our setting.

3.2 Medium-Run Analysis: Municipality-Level Long Differences

The establishment-level analysis captures how existing plants respond to annual temperature shocks, but does not answer how prolonged exposure to elevated temperatures reshapes local labor markets. If local economies adapt to annual temperature fluctuations, long-run effects may be attenuated. It is also feasible that responses in a given market are not driven by annual weather shocks, especially if such shocks are interpreted as idiosyncratic and may be reversed in the following years (Ponticelli et al., 2023). Conversely, if adaptation is limited or the shocks are persistent, climatic impacts can accumulate over time, leading to compounding and potentially more severe consequences.

To address this question, we shift to municipality-level analysis and adopt a long-differences approach, comparing the average number of *Heat Days* between the first and last decade of our panel (2000–2009 versus 2010–2019), as presented in Equation (3). This methodology follows recent empirical work studying medium- and long-run climatic effects, allowing us to examine how outcomes respond to persistent changes in climate relative to the short-run temperature shocks captured in our panel specification.[19] We investigate two sets of outcomes. The first captures adjustments in the local workforce: overall employment, measured by the total number of formal workers. The second captures adjustments along the exten-

[18] We do not have municipality yearly data on urbanization, so we use establishments and employment as proxies.

[19] See, for instance, Burke and Emerick (2016), Liu et al. (2023), and Ponticelli et al. (2023).

sive margin: the stock of establishments and the number of them entering and exiting. All variables are measured at the municipality level, and we also account for the number in each productivity category, which we now divide into quintiles instead of deciles.[20] Entry and exit are *flow* measures, which allow us to understand where the variations in stock come from. Examining these margins is essential, since aggregate employment figures alone cannot reveal whether climate change reshapes market composition by altering firm turnover dynamics. If prolonged heat exposure discourages entry or increases exit from productive establishments, climate change may slow productivity growth even if surviving firms adapt successfully.

$$\begin{aligned}\Delta lnY_{m,\overline{2010}s-\overline{2000}s} = \beta\, \Delta HeatDays_{m,\overline{2010}s-\overline{2000}s} + \delta\, \Delta Precip_{m,\overline{2010}s-\overline{2000}s} \\ + \alpha_g + \boldsymbol{\Omega}' \boldsymbol{X}_{m,t=2000} + \varepsilon_m\end{aligned} \tag{3}$$

The dependent variable $\Delta lnY_{m,\overline{2010}s-\overline{2000}s}$ is the difference (in natural logarithms) of a given outcome of interest between the average value observed in 2010–2019 and the average value in 2000–2009 for municipality m. We take the same differences for our treatment variable, $\Delta HeatDays_m$, and for the precipitation control. The term α_g controls for state fixed effects.

The identification assumption in Equation (3) relies on the exogeneity of the change in the average number of *Heat Days* between the first and the second decade. That is, the variation in the decennial average of *Heat Days* is uncorrelated with other local trends that might affect the outcomes. Our preferred specification includes state fixed effects, so that identification comes from within-state variation, mitigating concerns about unobservable trends operating over broad areas of the country. We further show that our main results are robust to replacing state fixed effects with more granular geographic units, such as mesoregions or microregions, which addresses the same concern at finer spatial scales. Yet, municipalities subject to increases in extreme heat may also differ in geographical characteristics that shape their subsequent development. We therefore add $\boldsymbol{X}_{m,t=2000}$, a vector of municipality-level characteristics in the baseline year (2000), which we collected from *Censo*. Specifically, we

[20] We use quintiles rather than deciles in the medium-run analysis because outcomes are also disaggregated by productivity and sector. At the decile level, many cells contain few or no establishments, yielding noisy or undefined variations. Quintiles preserve enough mass per cell, allowing us to retain a similar productivity ordering.

include as explanatory variables the shares of students with complete high school, rural population, and informal workers, as well as logs of per capita GDP and population.

Still, following the same reasoning as in our establishment-level specification, reverse causality caused by economically more dynamic municipalities warming up more could be a potential threat. In Table B.3, we regress $\Delta HeatDays$ on economic growth measures (growth in the stock of establishments and employment) between the two halves of the first decade (2000–04 to 2005–09) to analyze whether municipalities that were growing the most at the beginning of the period subsequently warmed the most.[21] Across all specifications, none of the estimated coefficients approaches conventional significance levels. Municipalities that were expanding faster in the early 2000s did not go on to experience differentially larger increases in extreme heat, supporting the orthogonality of this variable to pre-existing local dynamism.

4 Short-Run Effects

4.1 Establishment-Level Responses

We begin by analyzing how establishments adjust to annual extreme temperature shocks. We first document the results *without* the productivity interaction term of Equation (2). Coefficients reported in Table 2 indicate that, in general, neither establishment size nor closure propensity is significantly affected by increases in anomalous hot days. These null effects hold across all specifications, whether or not precipitation is controlled for and regardless of the combination of fixed effects. In contrast to studies documenting significant aggregate output losses from heat exposure at the country or industry level (Dell et al., 2014; Heal and Park, 2016; Acevedo et al., 2020; Somanathan et al., 2021; Rexer and Sharma, 2026), our establishment-level estimates indicate that these null aggregate effects likely reflect offsetting responses across heterogeneous establishments, rather than an absence of climatic impact.

[21] We use growth over the five-year sub-periods rather than over the full decade because growth from 2000–2009 temporally overlaps with the first half of the window over which $\Delta HeatDays$ is computed, meaning the "prior" outcome would itself be contaminated by the heat of the 2000s. Using the 2000–04 $\rightarrow$ 2005–09 transition approximates the ideal of predetermined local dynamism, albeit imperfectly, since heat during 2000–2009 also affects this growth.

Table 2: Aggregate Effects of *Heat Days* on Establishment Outcomes

Panel A: Dependent variable – Size						
Heat Days	−0.1299	0.1013	−0.1285	0.1036	0.1230	0.1259
	(0.0861)	(0.0928)	(0.0945)	(0.0900)	(0.0981)	(0.0953)
Precipitation			yes	yes		yes
Establishment FE	yes	yes	yes	yes	yes	yes
Industry x Year FE	yes		yes		yes	yes
Macroregion x Year FE		yes		yes	yes	yes
Observations	42,748,788	42,748,801	42,745,674	42,745,687	42,748,788	42,745,674
Adjusted R^2	0.8074	0.8053	0.8074	0.8053	0.8075	0.8075
Panel B: Dependent variable – Exit propensity						
Heat Days	−0.0209	−0.0022	−0.0228	−0.0075	−0.0074	−0.0069
	(0.0143)	(0.0141)	(0.0145)	(0.0172)	(0.0131)	(0.0139)
Precipitation			yes	yes		yes
Establishment FE	yes	yes	yes	yes	yes	yes
Industry x Year FE	yes		yes		yes	yes
Macroregion x Year FE		yes		yes	yes	yes
Observations	42,748,788	42,748,801	42,745,674	42,745,687	42,748,788	42,745,674
Adjusted R^2	0.1606	0.1412	0.1606	0.1412	0.1612	0.1612

Notes: Dependent variables are establishment size (Panel A) and exit propensity (Panel B). Coefficients and standard errors are multiplied by 1,000 to represent the effect of 10 *Heat Days* in percent for size and percentage points for exit. Standard errors are clustered at the municipality level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Motivated by these aggregate null effects, we then investigate how the yearly variation in extreme heat days produces heterogeneous effects on establishments' outcomes depending on their productivity level, shedding light on the potential *redistributive* effects triggered by these temperature shocks. Figure 3 reports the results on size. The negative response is sharply concentrated at the bottom of the productivity distribution, as plants in the first decile shrink by 0.70% following ten additional *Heat Days* in a year, an effect at least twice

as large in magnitude as any other decile. Establishments in the second and third deciles are essentially unresponsive, while those from the fourth to the ninth decile expand their workforce by between 0.24% and 0.34%, and the effect eventually fades away for the most productive ones.

The extensive margin tells a similar story. Figure 4 displays a nearly monotonic gradient that runs throughout the productivity distribution. Ten additional *Heat Days* increase the annual probability of closure by 0.26, 0.11, and 0.05 percentage points for plants in the first, second, and third deciles, respectively. Establishments in the middle of the distribution do not adjust along this margin, and from the sixth decile onward the sign reverses, with the most productive plants becoming 0.04 to 0.13 p.p. less likely to shut down. These magnitudes are best read against how frequently each group closes to begin with. The baseline exit rate in our full sample is 7.91%, being larger in the bottom deciles. The effect we estimate for the least productive plants amounts to a 2.44% increase in the annual closure risk of plants in this decile, while the ninth decile experiences a 1.87% reduction in its own, highlighting the economic magnitude of these impacts.

Figure 3: Effects on Establishment Size (in %) of 10 *Heat Days*

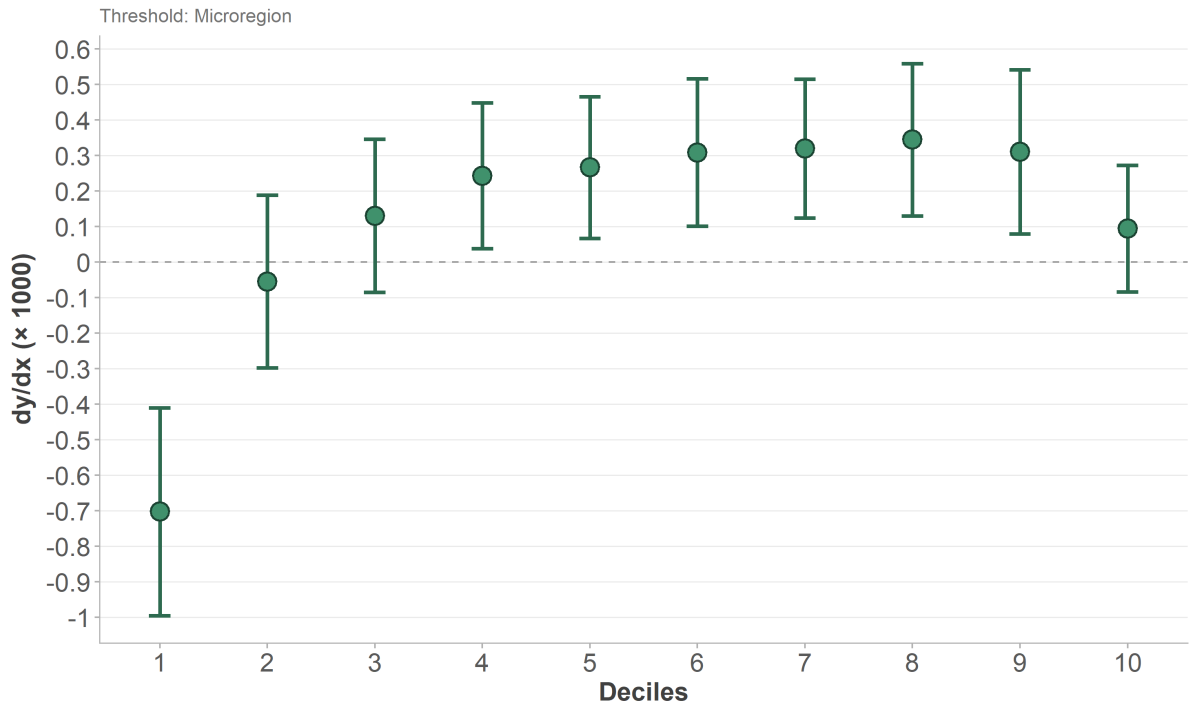
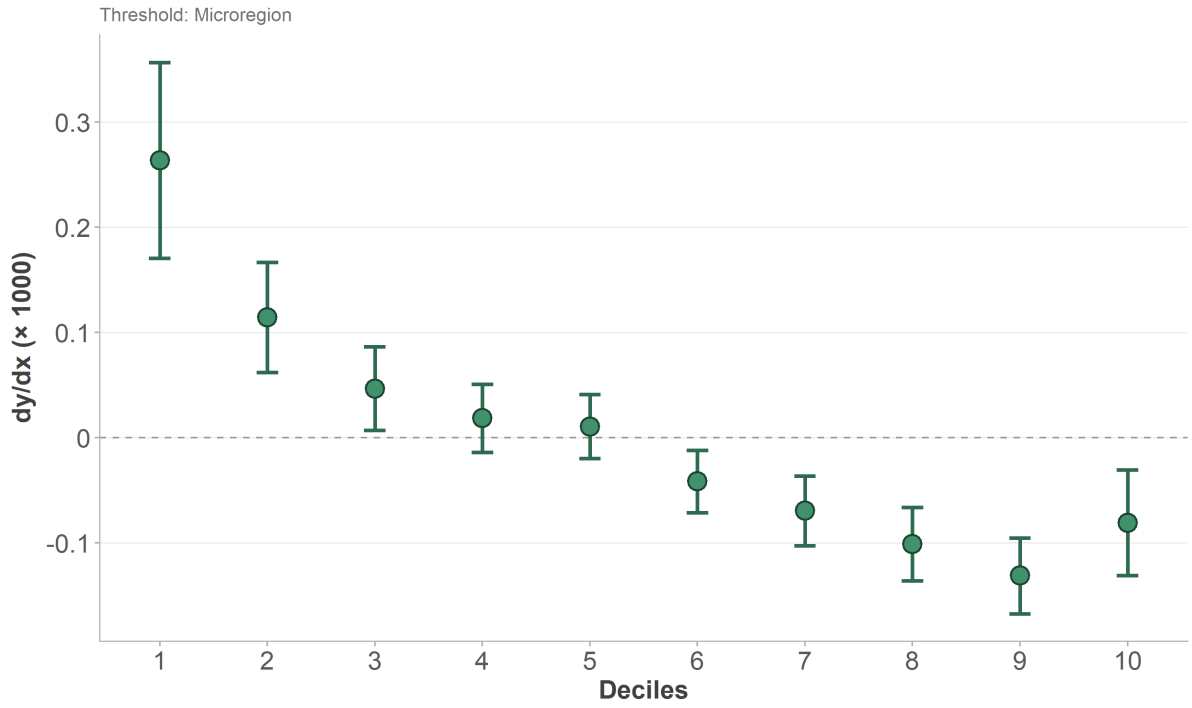


Figure 4: Effects on Establishment Exit Propensity (in p.p.) of 10 *Heat Days*



The two figures tell an equivalent story of how plants adjust. Least productive plants are the smallest in our sample, with a median of two employees (Table 1), leaving them little room to absorb a shock by downsizing. As a result, they contract sharply and close. Establishments higher in the distribution adjust in the opposite direction on both margins, expanding their payroll and surviving at higher rates.

If the driver of these redistributive effects is indeed the number of days with high temperatures, days with moderate temperatures should not affect our establishment outcomes. Figures B.1 and B.2 replace *Heat Days* with a placebo, defined as the annual count of *Mild Days* – those with maximum temperatures between the 45th and 55th percentiles. The coefficients display no systematic ordering across the distribution, and the sharp contrast between the tails disappears, confirming that the gradient we document is not an artifact of temperature variation as such.

These results are consistent with other findings in the literature, pointing to a short-run redistributive pattern in response to high temperatures (Tarsia, 2025). The concentration of losses at the bottom of the distribution is in line with the view that less productive units often

operate under tighter financial constraints, face limited access to credit, and are less likely to invest in adaptation or maintain adequate infrastructure – constraints that bind particularly hard in developing economies. On the other hand, more productive plants usually have more margins available to insulate themselves from the shock. Beyond this direct resilience, the pattern is also consistent with a reallocation channel: as heat-induced closures release workers from less productive establishments, their more productive counterparts may absorb part of this displaced labor, expanding employment and further reducing their own closure risk. In this interpretation, extreme heat accelerates the reallocation of production factors toward the upper tail of the productivity distribution.

Given that different economic sectors are likely to exhibit varying degrees of vulnerability to temperature shocks, it is important to explore sectoral heterogeneity. For instance, agricultural establishments may be particularly sensitive to extreme heat due to the direct negative impacts on crop yields (Deschênes and Greenstone, 2007; Burke and Emerick, 2016; Blanc and Schlenker, 2017), while the manufacturing sector employs a higher share of workers engaged in routine-manual tasks (Canon and Marifian, 2013; Marcolin and Squicciarini, 2016; Rocha, 2024), which tend to be more affected through the labor productivity channel.[22] Moreover, there is also evidence documenting the direct impact of abnormal temperatures on consumer demand, especially through its effects on industries in the non-tradable and consumer-oriented sectors, where the majority of the service establishments operate (Liu et al., 2023; Addoum et al., 2023; Jin et al., 2025). We therefore re-estimate Equation (2) separately for the agriculture, manufacturing, and services sectors. Figure 5 presents the results for size, while Figure 6 shows the findings related to exit propensity.

[22]In fact, previous literature found that labor productivity in the manufacturing sector decreases with high temperature shocks (Heal and Park, 2016; Somanathan et al., 2021; Costa et al., 2024).

Figure 5: Effects by Sector on Establishment Size (in %) of 10 *Heat Days*

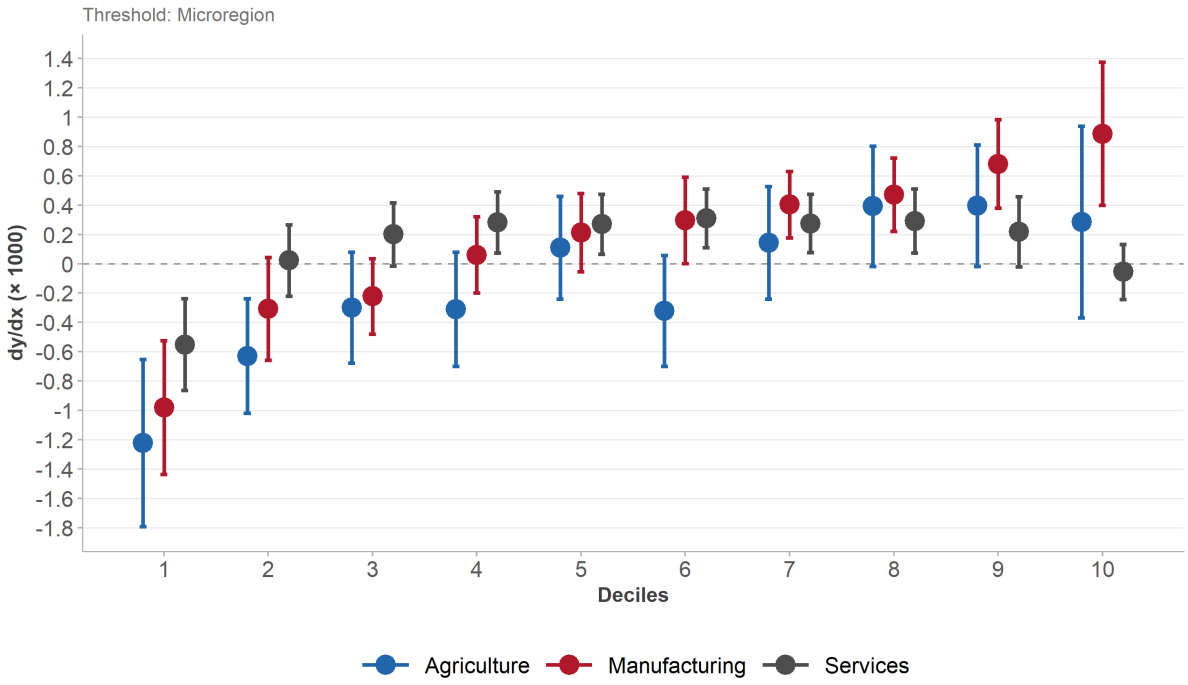
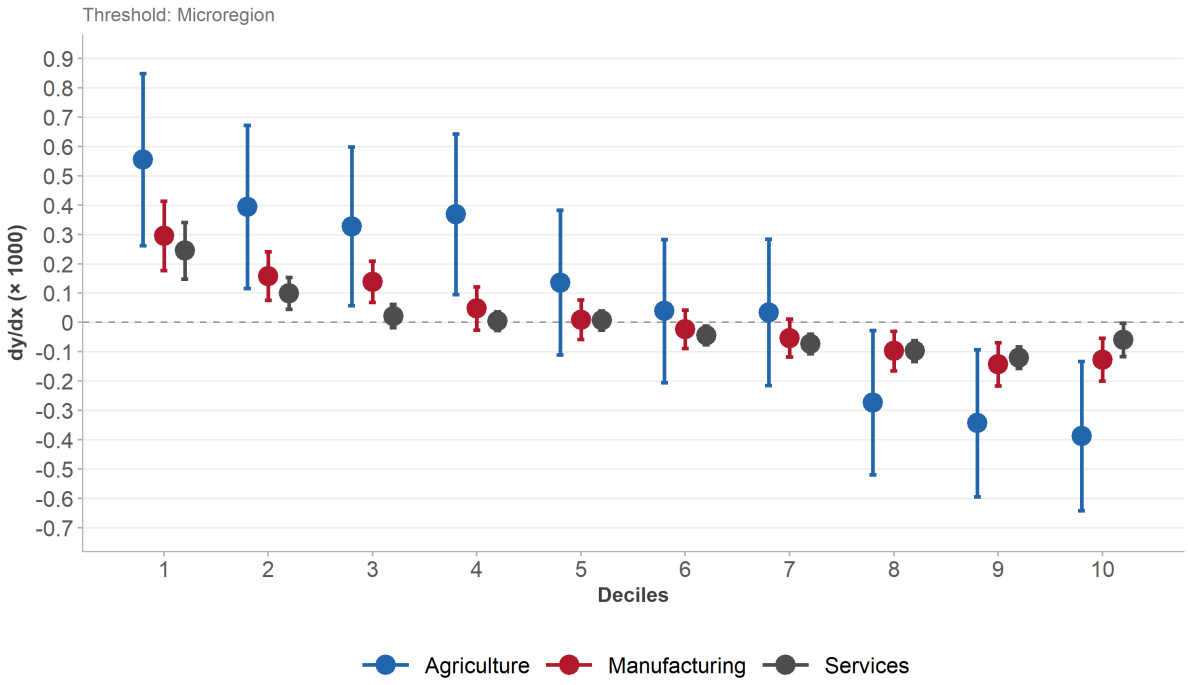


Figure 6: Effects by Sector on Establishment Exit Propensity (in p.p.) of 10 *Heat Days*



In agriculture, which represents the smallest share of the Brazilian formal workforce (see Table A.1), the least productive establishments significantly reduce their size, with contractions of 1.22% in the first decile and 0.63% in the second. However, these losses are no longer offset by robust increases in the middle and upper deciles of the distribution. Low-productivity agricultural plants are also more likely to close. Ten extra *Heat Days* raise the exit propensity of the first decile by 0.55 p.p., or 3.93% of its baseline rate, and the adverse effect extends up to the fourth decile (0.37 p.p., or 2.68%) – deeper into the distribution than in the other two sectors. Beneficial results appear only in the last three deciles and remain smaller, on average, than those observed at the bottom. Agriculture nonetheless displays the largest negative effects of any sector, which is unsurprising given how directly heat operates: extreme temperatures damage yields through the crop cycle itself, and the sector's workforce performs manual tasks outdoors, with little scope for shielding from ambient conditions.

The manufacturing sector presents particular characteristics, where the point estimates for the intensive margin are monotonic across the distribution. While first-decile establishments contract substantially (-0.98%), along with marginally significant decreases in the second and third deciles, the redistributive effects are stronger the more productive the manufacturing plant is, as from the seventh decile onward there is a growing expansion of workforce following additional *Heat Days*, reaching 0.89% at the top decile of the residual wage distribution. With respect to the extensive margin, plants up to the third decile become more likely to close, with larger magnitudes than in the aggregate results, reaching 0.30 p.p. among the least productive – a 2.97% increase over their baseline risk. As in agriculture, the heat-driven reduction in closure probability appears only in the top three deciles, and with smaller magnitudes than at the bottom of the distribution. The sharper redistributive gradient in manufacturing likely reflects the tighter competitive linkages within this sector. Manufacturing establishments compete more directly for customers and share local labor pools of workers with transferable sector-specific skills, so that heat-induced contractions at the bottom of the distribution simultaneously increase demand and labor supply that more productive plants can absorb at relatively low cost. In agriculture, by contrast, commoditized output markets and less specialized labor limit the scope for such reallocation.

As services encompass more than 83% of our sample, the coefficients' behavior across

the distribution for this sector is almost a mirror of the aggregate results, with *Heat Days* inducing a 0.55% size reduction in the bottom decile, and the expansions further up are modest, ranging from 0.28% to 0.31% between the fourth and eighth deciles. Notably, these gains do not extend to the top, as we see much noisier effects in the ninth and tenth deciles. A plausible reading is that reallocation operates strongly among plants that compete with one another: those in the middle of the distribution are close substitutes for the affected ones, while establishments at the very top serve distinct segments. This constraint should bind harder in services, where markets are local and non-tradable, than in manufacturing, where output is traded across regions, and the size gradient indeed rises all the way to the most productive plants. The same logic accounts for the attenuation of the pooled results: because services dominate the sample, the flat tenth-decile coefficient in Figure 3 reflects the absence of gains at the top of this sector rather than a general feature of the productivity gradient. Furthermore, adjustment along the extensive margin is also small here. The point estimates in Figure 6 are smaller in magnitude than in the other two sectors, both for the losers and for the winners, which suggests that consumer demand is not the dominant channel at work: were it so, one would expect the response of this subsample to be stronger rather than weaker.

In short, our sectoral disaggregation adds two things to the picture. First, the *Heat Days* effects continue to line up with productivity within each sector: in agriculture, manufacturing, and services alike, the least productive plants are the main losers, while those higher up perform better. The redistributive pattern is therefore a matter of heterogeneity across plants rather than of the relative weight of sectors in the economy – a distinction the pooled specification alone cannot make. Second, what varies is the shape and the intensity of that ordering. This variation is itself suggestive about mechanisms: responses are strongest in the two sectors where extreme heat bears directly on production and labor productivity (agriculture and manufacturing), and weakest in the one where it would operate mainly through consumer demand (services).

4.2 Adjustment Margins: Workforce Composition and Trade

We also examine whether the reallocation of employment changes establishments' workforce composition. These changes can arise for several reasons that are not mutually exclusive. First, more productive plants may increase the share of more educated workers as they adapt their production processes to the increase in high temperatures. If firms raise the demand for qualified labor to mitigate the negative effects of heat, for instance by implementing new equipment or practices, the remaining workforce will be more skill-intensive, given the complementarity between capital and skilled labor. Second, composition may change through shifts in labor supply: establishments that contract or close increase the supply of workers in the local labor market, allowing more productive plants to hire from a larger pool and possibly at lower wages. Moreover, workers with better outside options may also sort toward high-productivity establishments. To test how *Heat Days* changes establishments' workforce composition, we compute, for each establishment, the number of college-graduate and non-graduate workers, as well as the share of graduate workers, and use them as outcomes in Equation (2).

Figure 7: Effects on the Level of Graduate and non-Graduate Workers (in %) of 10 *Heat Days*

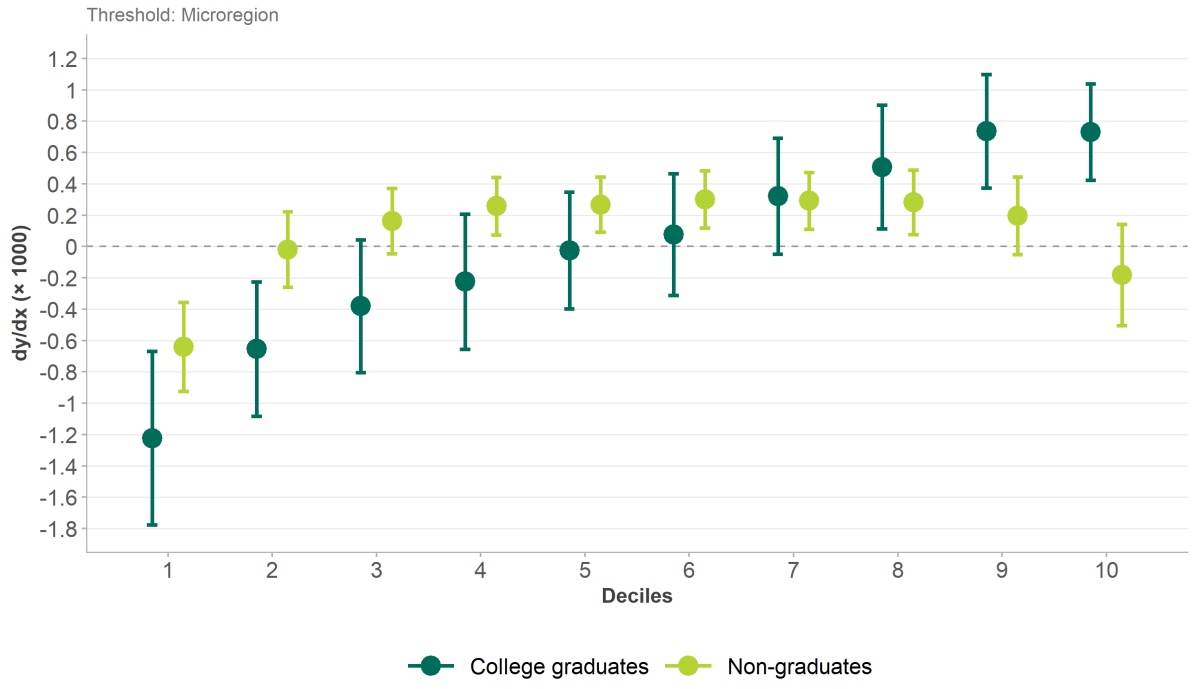


Figure 8: Effects on the Share of Graduate Workers (in p.p.) of 10 *Heat Days*

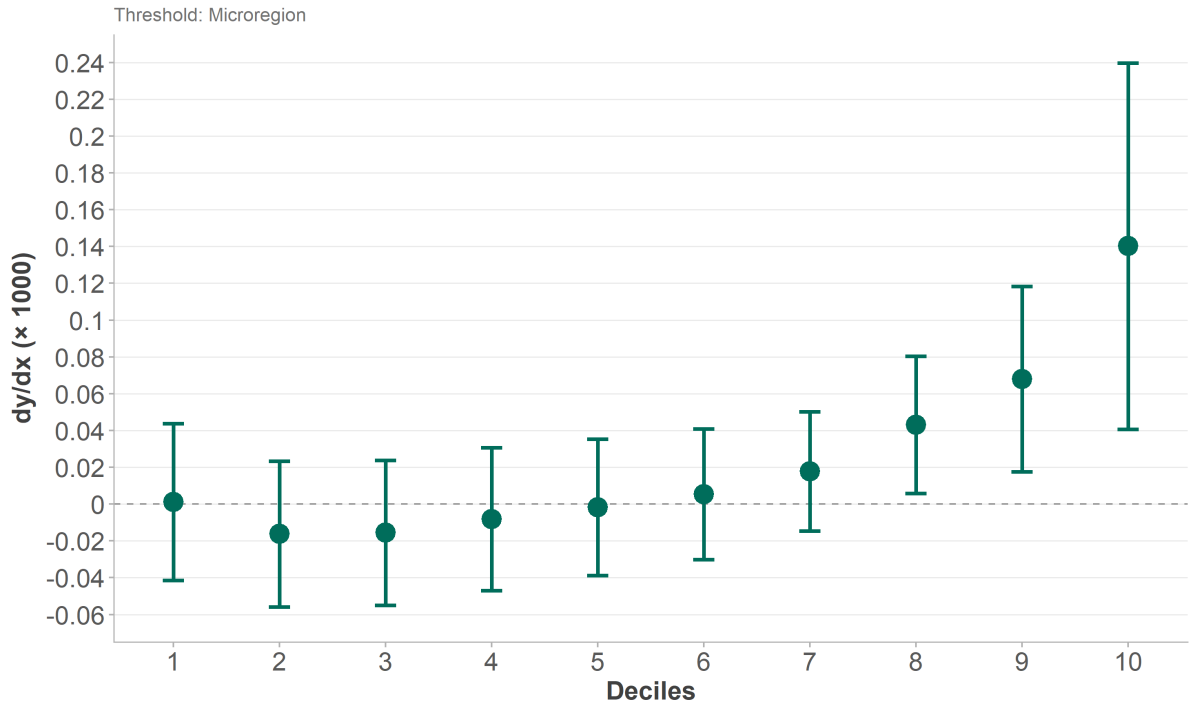


Figure 7 reports the effect of *Heat Days* on the log number of graduate and non-graduate workers, and Figure 8 reports the effect on the share of graduate workers. Since displaced workers are mostly non-graduates, absorbing them would lower an establishment's graduate share. Instead, a rise indicates upskilling or sorting of educated workers toward productive plants. Our findings reveal that the least productive plants lose workers of both types, while plants between the fourth and seventh deciles expand mainly through non-graduates, with no change in their graduate share. From the eighth decile onward, however, establishments significantly increase graduate employment while the gains in non-graduate employment start to fade out, so the share of graduates rises in the upper deciles. For the most productive plants, ten additional *Heat Days* raise this share by 0.14 p.p., or 0.83% of its baseline mean. Since the displaced pool is less educated than the workforce of top plants, absorption alone is unlikely to explain this result. Although small in magnitude, the increase is consistent with the most productive establishments shifting toward more skilled labor, which may reflect their greater capacity to adapt in the short run. Figure A.3 and Figure A.4 repeat the exercise using a measure of qualification based on occupation rather than schooling.[23] The responses of high- and low-skill employment are similar to those obtained with schooling, but the share of high-skill workers responds only marginally at the top (0.09 p.p., or 0.41% of its baseline mean), which suggests that the shift occurs largely within occupational categories.

Nonetheless, the increase in graduate employment at the top could reflect better-educated workers sorting toward more productive plants, rather than a change in how these plants operate. Sorting, however, is unlikely to predict that plants adjust in other margins. Figure A.5 shows that *Heat Days* also raise the probability that an establishment imports, with positive and significant effects from the fifth decile onward. In the top deciles, where the graduate share rises, plants thus also become more likely to buy from abroad, which is consistent with adaptation through imported equipment or inputs operated by a more educated workforce. We observe, however, only whether a plant imports, not what it imports, and growth or sorting may also contribute to this pattern. These results should therefore be read only as suggestive evidence of a differential adaptive capacity.

[23] High-skilled occupation workers comprise managers, professionals, and technicians. Low-skilled workers include clerical support, service and sales workers, agricultural laborers, craft and related trades workers, plant and machine operators, and elementary occupations.

5 Medium-Run Effects

5.1 Aggregate and Distributional Effects

We now turn to the municipality-level long-differences analysis. As outlined previously, the redistributive pattern documented in the short run may be reinforced over time if adaptation is limited, particularly among the least productive establishments, or it may instead reflect a transitory market adjustment to extreme heat. In the latter case, longer horizons may point to distinct conclusions. We study the impacts on four outcomes, all aggregated at the municipality level and expressed as log differences between the 2010–2019 and the 2000–2009 averages: (i) employment, the total number of formal workers; (ii) establishments, the number of formal plants; (iii) entry, the number of plants that did not operate in $t-1$ but did in t; (iv) exit, the number of plants that operated in t but no longer did in $t+1$. Table 3 reports the estimates from Equation (3). Our preferred specification includes state fixed effects (Panel A), but we show that our results are robust to replacing them with more granular geographic controls, namely mesoregion and microregion fixed effects (Panels B and C).

Table 3: Medium-Run Effects of *Heat Days* Across Municipalities

	Employment	Establishments	Entry	Exit
Panel A: State FE				
Δ Heat Days	-2.4459^{*}	-2.0961^{**}	-3.4187^{***}	0.1233
	(1.4677)	(0.8206)	(0.9186)	(0.9813)
Panel B: Mesoregion FE				
Δ Heat Days	-1.7087	-2.7658^{***}	-3.8270^{***}	0.3380
	(1.7155)	(0.9593)	(1.0475)	(1.1655)
Panel C: Microregion FE				
Δ Heat Days	-1.8575	-2.8051^{***}	-4.0218^{***}	-0.8279
	(1.8257)	(0.9625)	(1.0951)	(1.2094)
Observations	5,489	5,489	5,476	5,427

Notes: The dependent variable in each column is the difference (in natural logarithm) of the outcome between the averages of 2010–2019 and 2000–2009 at the municipality level. Establishments is the stock of active plants; entry is the number of establishments that did not exist in $t-1$ but existed in t, and exit is the number that existed in t but not in $t+1$. Each panel reports the same set of regressions varying only the geographic fixed effect. Coefficients are multiplied by 1,000, so that they represent the effect of a ten-day increase in the decennial average of *Heat Days*, in percent. All specifications control for the change in average precipitation between the two decades and for municipality-level characteristics at the baseline year (2000). Standard errors are clustered at the municipality level. * p<0.10, ** p<0.05, *** p<0.01.

Prolonged exposure to extreme heat diminishes the local establishment base. A ten-day increase in the decennial average number of *Heat Days* is associated with a contraction of 2.10% to 2.81% in the stock of active plants, an effect that is robust to our geographic fixed effects choice. Crucially, this contraction does not operate through firms shutting down: the exit coefficients are small and statistically insignificant. What declines instead is firm entry. The number of new establishments falls by 3.42% to 4.02%, the largest estimated effects in the table. Furthermore, employment moves in the same direction, but the effect is considerably smaller: the point estimates are consistently negative, yet only marginally significant under state fixed effects and indistinguishable from zero elsewhere, suggesting that, if extreme heat reduces municipal employment over a decade, the effect is small compared to the adjustment

we observe in the number of establishments.

In the short-run results, the aggregate effect of *Heat Days* was null on both margins, and a result of offsetting movements across the distribution. In the medium run, the aggregate effect is itself negative, indicating that prolonged exposure to extreme heat is damaging to local economies rather than merely redistributive within them. This is consistent with previous research on the medium- and long-run consequences of global warming in local economies (e.g., Ponticelli et al. (2023) and Albert et al. (2024)). It also suggests that the short-run reallocation is not entirely absorbed by the surviving plants. The natural next question is therefore who bears these losses, and whether the productivity gradient observed in the short-run survives at longer horizons. We take a look at this in Table 4 below.

The first column shows that the marginal aggregate employment decline documented in Table 3 cannot be attributed to establishments in any particular part of the productivity distribution. The point estimates are negative throughout and largest among the most productive plants, but none is statistically distinguishable from zero. The stock of establishments behaves differently. Here the contraction is concentrated at the top: municipalities exposed to a ten-day rise in their decennial average of *Heat Days* saw the number of plants in the highest productivity quintile fall by 2.64%, while the remaining quintiles display smaller and imprecise coefficients.

Table 4: Medium-Run Effects of *Heat Days* Across Municipalities, by Productivity Level

	Employment	Establishments	Entry	Exit
Quintile 1	−0.2162	−0.2113	−2.4718*	1.8661
	(1.9508)	(1.3229)	(1.4200)	(1.5048)
Quintile 2	−0.4686	−1.1266	−2.0805	0.6666
	(1.6974)	(1.1060)	(1.3321)	(1.4280)
Quintile 3	−1.6386	−0.7960	−0.6999	0.2889
	(1.7460)	(1.0676)	(1.2444)	(1.4326)
Quintile 4	−0.6592	−1.3423	−3.1971**	−2.3919
	(1.8688)	(1.0216)	(1.4299)	(1.4922)
Quintile 5	−2.1540	−2.6435**	−4.1221***	0.8084
	(2.4917)	(1.1241)	(1.4711)	(1.6134)
Observations	[5,379; 5,418]	[5,379; 5,418]	[5,156; 5,235]	[4,843; 5,059]

Notes: The dependent variable in each column is the difference (in natural logarithm) of the outcome between the averages of 2010–2019 and 2000–2009 at the municipality level. Establishments are ranked into quintiles of the productivity proxy within their local labor market. Coefficients are multiplied by 1,000, so that they represent the effect of a ten-day increase in the decennial average of *Heat Days*, in percent. All specifications include state fixed effects, a control for the change in average precipitation between the two decades, and for municipality-level characteristics at the baseline year (2000). Standard errors are clustered at the municipality level. * p<0.10, ** p<0.05, *** p<0.01.

The aggregate results indicate that the shrinking establishment base reflects a decline in firm entry rather than an acceleration in firm exit. Disaggregating by productivity sharpens this finding considerably: what $\Delta HeatDays$ suppresses is the creation of more productive plants. An increase in extreme heat reduces the volume of new fourth quintile establishments by 3.20%, and by 4.12% for those in the fifth quintile. The bottom quintile also records a decline, but one that is smaller in magnitude and only marginally significant, so that the entry response strengthens as one moves up the distribution. Crucially, this decline in high-productivity entry is not offset by an accommodating pattern on the exit margin, nor by higher exits on the number of low-productivity plants. Such a pattern may imply a dampening effect on aggregate productivity growth, as prolonged climate shocks reduce the entrance of high-productivity plants without proportionally reducing the quantity of

less productive ones, a process that can hinder the reallocation dynamics that typically drive efficiency gains in local economies, amplifying the long-run costs of climate change. A related result appears in Cascarano et al. (2025), who find that in Italy the reduction in entry associated with additional hot days is driven by smaller firms. That the creation of plants falls on the most productive plants in Brazil (a LMIC), rather than the smallest ones as in a high-income economy country, is consistent with the broader premise of this paper: the margins along which firms adjust to heat may differ systematically between developed and middle-income countries.

We also investigate whether these patterns are common to all sectors or concentrated in a few. Figure A.7 shows that the contraction in the establishment stock is concentrated in agriculture, where the coefficient is negative in every quintile and reaches 4.09% at the top. Estimates for this sector are imprecise, since many municipalities have no formal agricultural plants in a given quintile. The entry results in Figure A.8 are also informative: suppressed creation of productive plants is driven by manufacturing and agriculture, with both turning significantly negative from the third quintile onward and reaching declines of 4.51% and 8.30%, respectively, at the top, while services show no such gradient. Employment and exit, in Figure A.6 and Figure A.9, remain weak once disaggregated, confirming that the null aggregate responses on these margins are not the product of offsetting sectoral movements.

5.2 Structural Consequences: Local Market Concentration

We now turn to the consequences of these dynamics for the structure of LLMs, guided by a simple question: does sustained exposure to extreme heat make local markets more concentrated? Two findings motivate this question. In the short run, *Heat Days* reallocate activity from less productive to more productive establishments, which appear to be better positioned to absorb the shock, with greater capability to adapt in the short run. Over the medium run, decennial warming reduces the number of active establishments, driven primarily by weaker entry of high-productivity plants. Together, these patterns suggest that local markets may be growing more concentrated.

To investigate this, we use two versions of the Herfindahl–Hirschman Index (HHI) as outcomes in Equation (3). Market shares are measured in terms of employment, so both

indices capture employment concentration in local markets. The first is an aggregate HHI, computed across all establishments in the municipality, which reflects both concentration within industries and shifts of employment across them. The second is a within-industry HHI, computed within each municipality–industry cell and aggregated to the municipality as an employment-weighted average across industries.[24] Columns (1) and (2) of Table 5 report the results. All estimates are positive, indicating that sustained exposure to extreme heat raises local labor market concentration over the medium run. The aggregate HHI is sensitive to the choice of geographic fixed effects, while the within-industry HHI, in contrast, is precisely estimated across all specifications, with a ten-day increase in the decennial average of *Heat Days* raising concentration by 1.70% to 3.03%. Taken together, these results suggest that the short-run reallocation toward more productive establishments may accumulate over time and go hand in hand with greater local concentration of employment and weaker entry of new establishments, the main margin of medium-run adjustment.

[24]Industries are defined by CNAE sections (21 industries).

Table 5: Medium-Run Effects of *Heat Days* on Local Market Concentration

	HHI		% Employment in Q5	
	Aggregate	Industrial	Incumbents	All
	(1)	(2)	(3)	(4)
Panel A: State FE				
Δ Heat Days	0.9658	1.7014**	0.4287*	−0.3248
	(1.2950)	(0.8360)	(0.2453)	(0.1996)
Panel B: Mesoregion FE				
Δ Heat Days	3.5710**	3.0332***	0.6916**	−0.0127
	(1.4757)	(0.9575)	(0.2827)	(0.2237)
Panel C: Microregion FE				
Δ Heat Days	3.5820**	2.6854***	0.3927	−0.0322
	(1.5406)	(1.0109)	(0.3136)	(0.2300)
Observations	5,489	5,489	5,489	5,506

Notes: The dependent variable in each column is the difference in the outcome between the averages of 2010–2019 and 2000–2009 at the municipality level. Columns (1) and (2) are log differences of the Herfindahl-Hirschman Index (HHI) of employment shares. The aggregate HHI (Column 1) is computed across all establishments in the municipality. The industrial HHI (Column 2) is computed within each municipality–industry cell (CNAE Section) and aggregated to the municipality level as an employment-weighted average across industries. Column (3) is the change in the share of total municipal employment held by establishments already operating at baseline year (incumbents) in the top productivity quintile (Q5), defined in that year. Column (4) is the change in the share of total municipal employment held by all Q5 establishments. Each panel reports the same set of regressions varying only the geographic fixed effect. Coefficients are multiplied by 1,000, so that they represent the effect of a ten-day increase in the decennial average of *Heat Days* – in percent for Columns (1) and (2), and in percentage points for Columns (3) and (4). All specifications control for the change in average precipitation between the two decades and for municipality-level characteristics at the baseline year (2000). Standard errors are clustered at the municipality level. * p<0.10, ** p<0.05, *** p<0.01.

If global warming shifts employment toward high-productivity establishments, their share of local employment would be expected to rise over the medium run. However, if warming is also associated with weaker entry of new establishments, the two patterns may partly offset each other. Column (3) restricts the employment share of fifth-quintile establishments to incumbents, defined as top-quintile establishments already operating in the baseline year,

2000, with their productivity quintile fixed at that year. The share held by high-productivity incumbents presents a rise, although slightly noisy, by 0.39 to 0.69 percentage points following a ten-day increase in the decennial average of *Heat Days*. Column (4), in contrast, uses all fifth-quintile establishments as the outcome, and shows that their aggregate employment share is unchanged across decades.[25] This net-zero effect suggests that the market is becoming more concentrated among incumbents. Existing high-productivity establishments gain employment, but fewer new ones enter. As a result, the top quintile's share of local employment stays roughly the same, while employment within it becomes more concentrated in the firms already there.

6 Conclusions

Global warming not only raises long-run average temperatures but also increases the frequency and intensity of short-run heat shocks. This paper combines climate data with formal establishment records spanning two decades in Brazil to examine how rising temperatures affect establishment performance and labor market dynamics. We employ a fixed-effects model where the yearly number of extreme heat days in a municipality is plausibly exogenous to analyze the short-run effects of high temperatures. In the medium run, we adopt a long differences approach to evaluate how climate change reshapes local economies.

Our findings reveal that, rather than uniformly reducing establishment performance, heat shocks trigger redistributive effects across productivity levels. In the short run, less productive establishments are disproportionately harmed by days of extreme heat, experiencing employment contractions and reduced survival rates. In contrast, the most productive plants appear to benefit from these dynamics, expanding in size and exhibiting greater resilience to closure. This indicates a heat-induced reallocation pattern that possibly operates along with a greater short-run adaptive capacity by high-productivity plants, which increase their share of more qualified workers and their probability of becoming importers. These winners-and-losers dynamics are stronger in manufacturing, where the gradient is monotonic across the

[25]The share of employment in other quintiles displays null results, so we omit those from the table for better readability.

whole distribution, while in agriculture the negative effects for less productive plants are not offset by gains at the top, and services display the smallest responses on both margins. Our medium-run analysis reveals that prolonged exposure to elevated temperatures reduces the number of establishments at the municipality level. We find that this contraction is mainly driven by a decrease in the entry of high-productivity establishments, while the exit margin remains unaffected. Such results are mainly driven by agriculture and manufacturing, while the service sector is not affected in the medium run. This entry reduction is accompanied by rising employment concentration within local industries, driven by the consolidation of high-productivity incumbents rather than by any expansion of high-productivity establishments as a group.

Taken together, these results highlight that the economic costs of climate change extend beyond aggregate output losses. In the short run, heat shocks accelerate the reallocation of resources toward more productive establishments. Over the medium run, however, this reallocation appears to compound into a more concentrated market structure, in which incumbent plants consolidate their position while weaker entry limits the emergence of new competitors. Because the medium-run contraction operates through establishments that are never created rather than through those that close, adaptation policies conditioned on incumbency address a margin along which we find limited damage. Instruments acting on the entry margin, such as lowering the sunk costs of setting up climate-resilient operations, are more likely to preserve the reallocation dynamics that sustain productivity growth.

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Appendix A: Additional Figures and Tables

Figure A.1: Decennial Growth (%) in the Number of *Heat Days* by Municipality

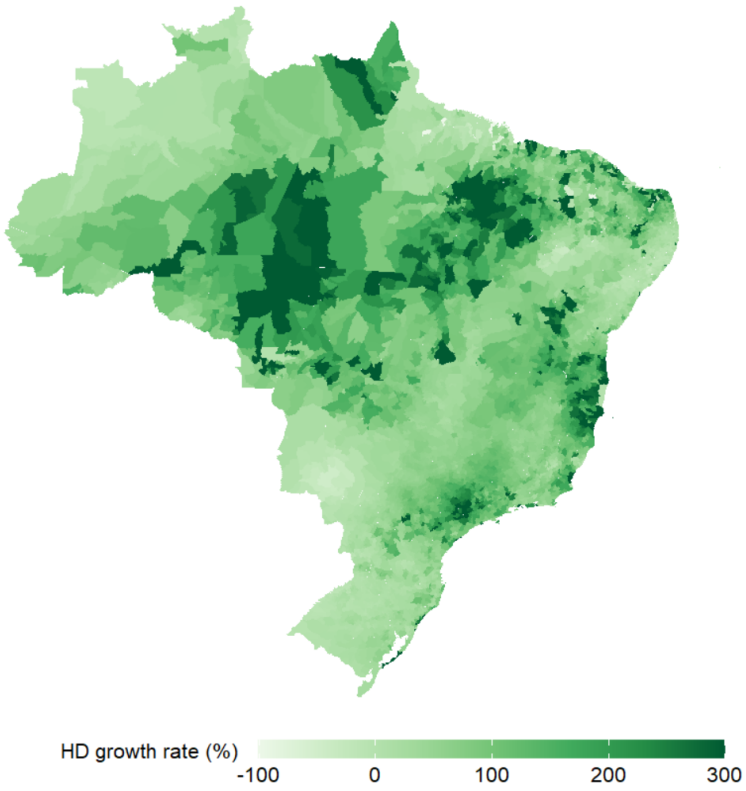


Figure A.2: Average Number of *Heat Days* by Region

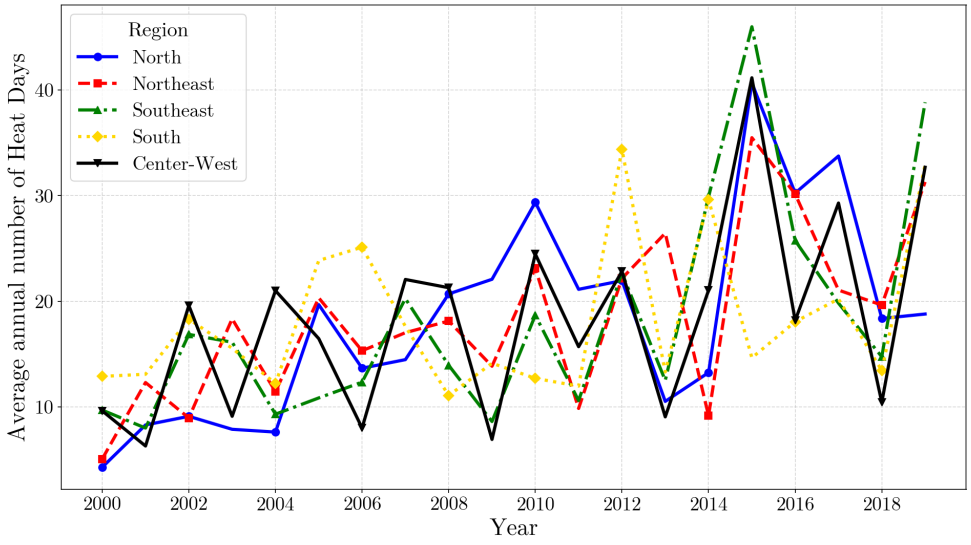


Table A.1: Shares of Workforce and Skill Composition by Sector and Productivity Level

Sectors	Plants	Workforce	High-Skill (Occupation)	Graduate
Agriculture				
Total	1.99%	2.22%	6.11%	3.31%
1st decile	10.03%	3.64%	5.29%	1.29%
5th decile	9.99%	7.04%	3.80%	1.21%
10th decile	9.97%	22.58%	11.04%	7.81%
Manufacturing				
Total	14.89%	31.71%	12.69%	7.42%
1st decile	9.94%	2.66%	9.85%	2.50%
5th decile	9.96%	5.91%	7.23%	2.96%
10th decile	10.31%	31.57%	20.32%	14.44%
Services				
Total	83.12%	66.08%	17.64%	9.98%
1st decile	10.02%	4.26%	24.73%	7.32%
5th decile	10.00%	8.68%	11.95%	5.09%
10th decile	9.93%	19.11%	29.35%	23.30%

Notes: Economic sectors are classified according to the two-digit CNAE (National Classification of Economic Activities) codes. We classify agriculture as CNAEs 1-3, manufacturing as CNAEs 5-42, and services as CNAEs 43-99. Workers are classified into two skill groups based on the Brazilian Classification of Occupations (CBO 2002): High-skilled occupation workers comprise managers, senior officials, professionals, and technicians (CBO 2002 Major Groups 1–3). Low-skilled workers include clerical support, service and sales workers, agricultural laborers, craft and related trades workers, plant and machine operators, and elementary occupations (CBO 2002 Major Groups 4–9). Graduate workers are those with a tertiary education degree.

Figure A.3: Effects on the Level of High-Skill and Low-Skill Workers (in %) of 10 *Heat Days*

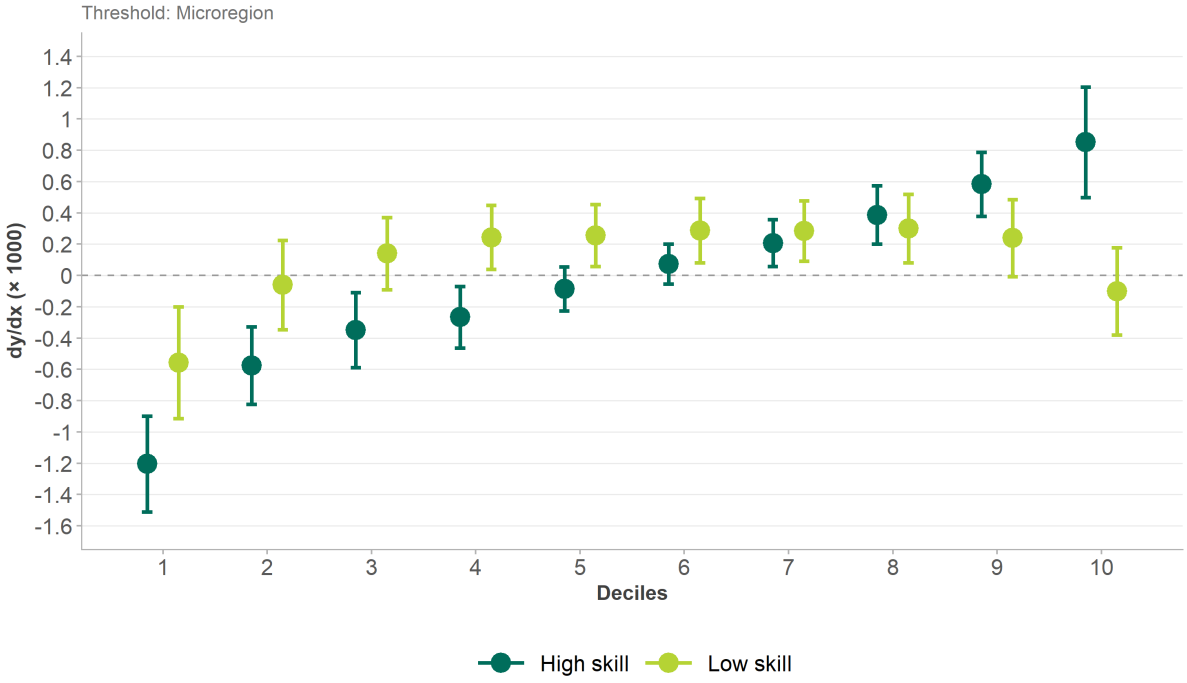


Figure A.4: Effects on the Share of High-Skill Workers (in p.p.) of 10 *Heat Days*

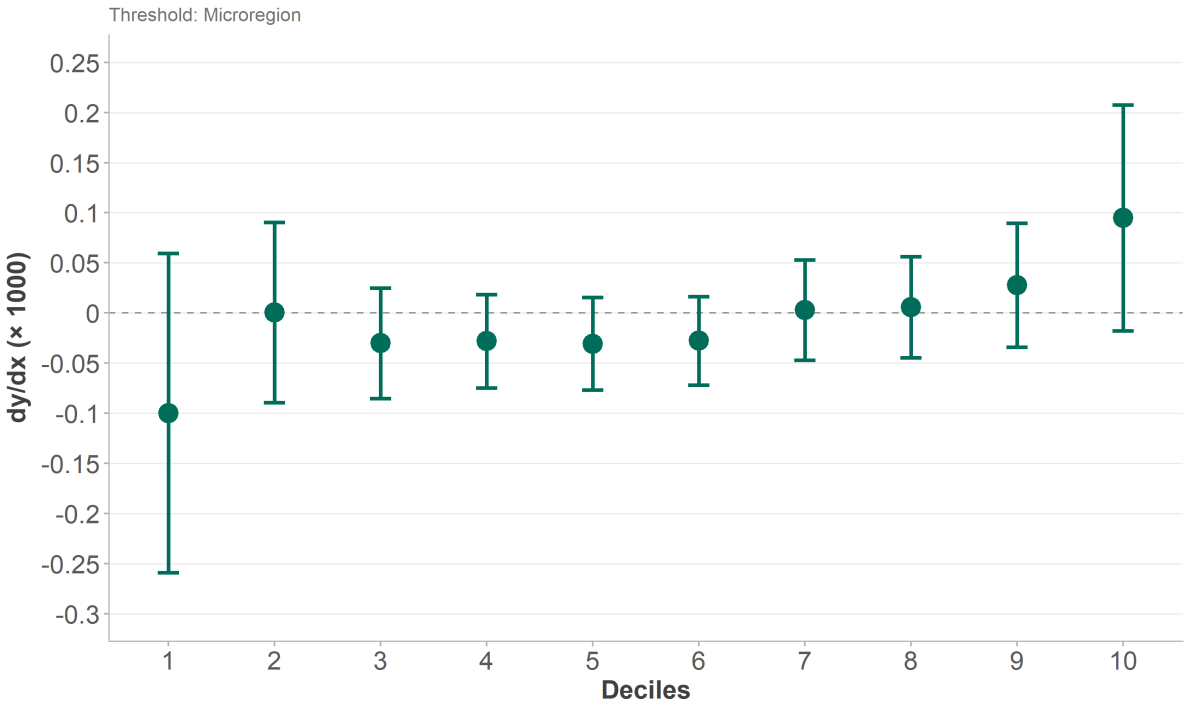


Figure A.5: Effects on the Import Probability (in p.p.) of 10 *Heat Days*

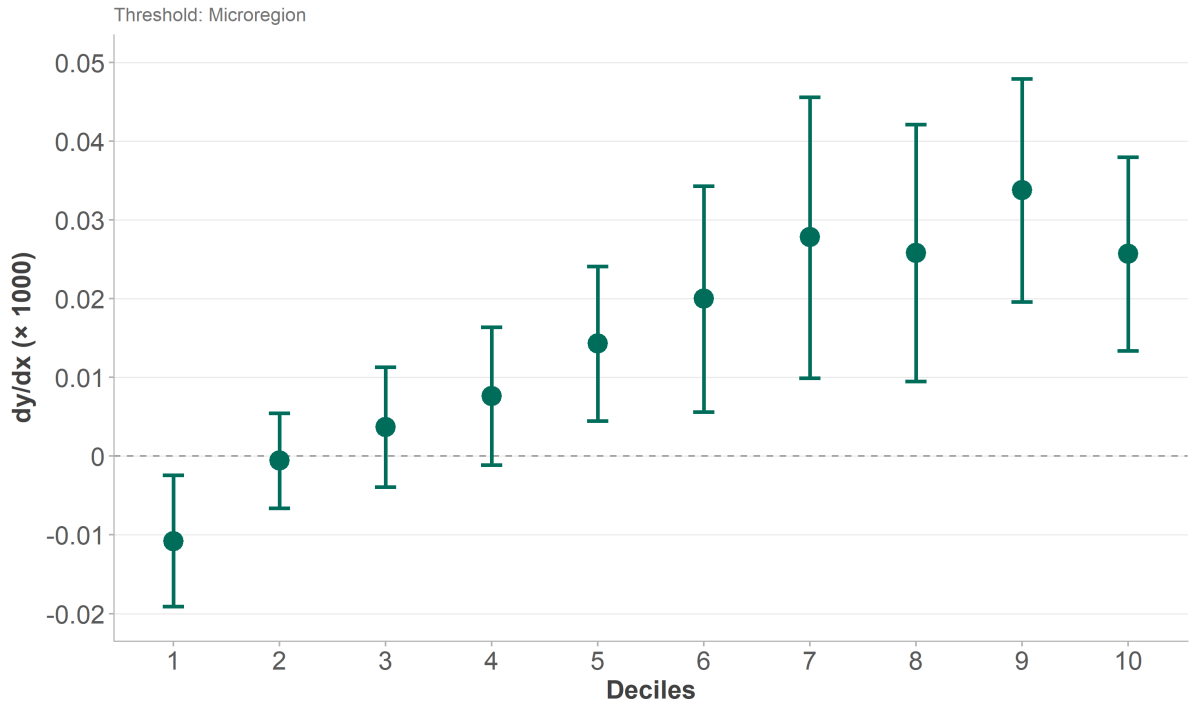


Figure A.6: Medium-Run Effects on Employment (in %) of 10 *Heat Days* Across Municipalities, by Productivity Level and Sector

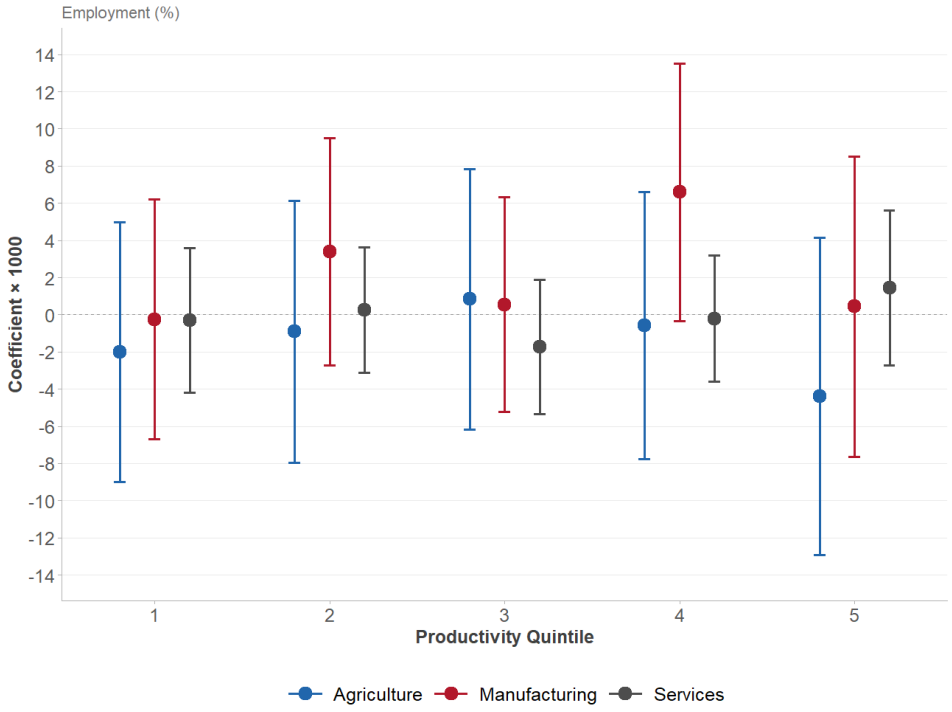


Figure A.7: Medium-Run Effects on Establishments Stock (in %) of 10 *Heat Days* Across Municipalities, by Productivity Level and Sector

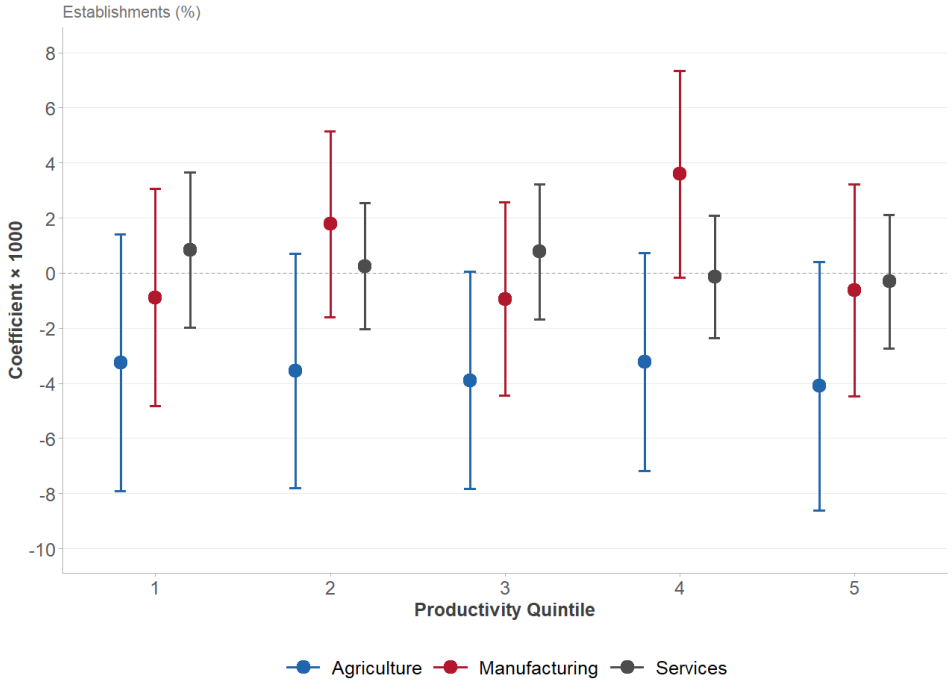


Figure A.8: Medium-Run Effects on Establishment Entry (in %) of 10 *Heat Days* Across Municipalities, by Productivity Level and Sector

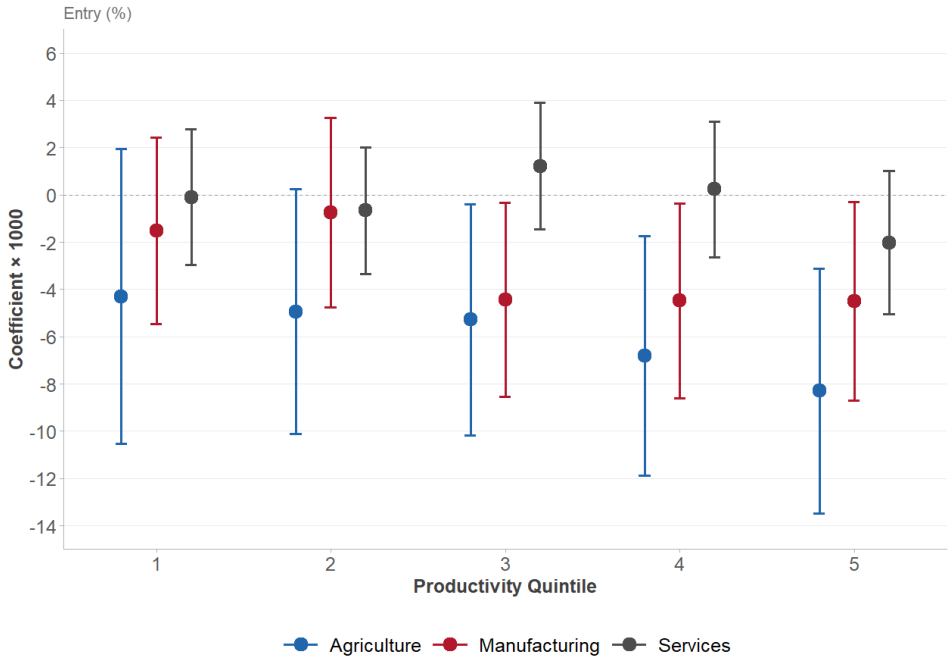
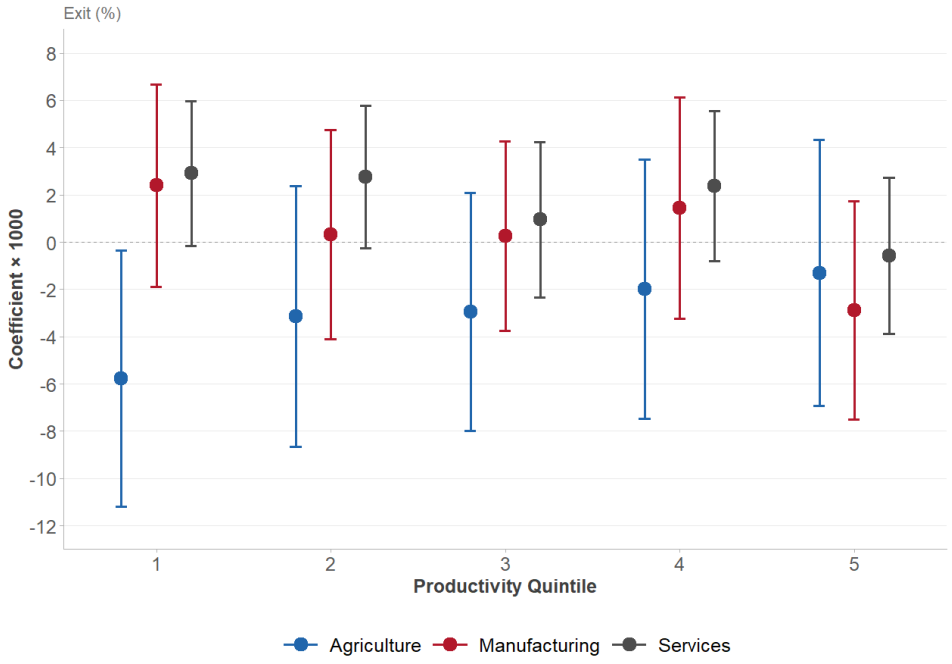


Figure A.9: Medium-Run Effects on Establishment Exit (in %) of 10 *Heat Days* Across Municipalities, by Productivity Level and Sector



Appendix B: Identification Tests

Table B.1: Effect of 10 *Heat Days* on Establishment Productivity and Decile Mobility

	Residual ($\hat{\mu}$)	**Distance**	**Change**	**Up**	**Down**
Panel A: Overall					
	−0.0015**	0.0014	−0.0358	0.0158	−0.0516
	(0.0006)	(0.0026)	(0.0319)	(0.0718)	(0.0637)
Panel B: By Productivity Decile					
1st decile	−0.0038***	0.0245***	0.4177***	0.5623***	—
	(0.0011)	(0.0039)	(0.0668)	(0.0822)	—
2nd decile	0.0008	0.0106***	−0.0215	0.3054***	−0.4569***
	(0.0005)	(0.0037)	(0.0426)	(0.0883)	(0.0830)
3rd decile	0.0013***	0.0030	−0.2095***	0.0488	−0.3192***
	(0.0004)	(0.0040)	(0.0435)	(0.1056)	(0.1097)
4th decile	0.0015***	−0.0031	−0.2601***	−0.1414	−0.1649
	(0.0004)	(0.0040)	(0.0482)	(0.1113)	(0.1164)
5th decile	0.0015***	−0.0060	−0.2266***	−0.2007*	−0.0652
	(0.0003)	(0.0046)	(0.0482)	(0.1164)	(0.1144)
6th decile	0.0015***	−0.0098**	−0.1980***	−0.2789**	0.0363
	(0.0003)	(0.0047)	(0.0475)	(0.1102)	(0.1012)
7th decile	0.0012***	−0.0145**	−0.1100***	−0.3678***	0.2045*
	(0.0004)	(0.0057)	(0.0376)	(0.1142)	(0.1051)
8th decile	0.0006	−0.0217***	−0.1317***	−0.5380***	0.3413***
	(0.0004)	(0.0066)	(0.0389)	(0.1213)	(0.1142)
9th decile	−0.0006	−0.0271***	0.0152	−0.6789***	0.5665***
	(0.0006)	(0.0069)	(0.0510)	(0.1352)	(0.1237)
10th decile	−0.0062***	−0.0306***	0.3996***	—	0.8839***
	(0.0013)	(0.0062)	(0.0798)	—	(0.1222)
Observations	42,691,841	33,654,880	33,654,880	30,044,978	30,401,207

Notes: Coefficients represent the effect of 10 additional *Heat Days* on the outcomes. All coefficients are obtained via Equation (2). Panel A reports the coefficient from the baseline specification for each outcome, without the productivity interaction term. Panel B reports marginal effects of *Heat Days* by productivity decile. *Distance* represents the cardinal change in productivity decile ranking between t and $t+1$. *Change*, *Up*, and *Down* are dummies indicating whether the plant changed, moved up, or moved down deciles between t and $t+1$, and the coefficients are multiplied by 100 to represent the effect in percentage points.

Table B.2: Reverse Causality: Effect of Realized Economic Activity on *Heat Days*

	Heat Days in t					
	(1)	(2)	(3)	(4)	(5)	(6)
Economic activity						
Δ ln (Employment)$_{t-1}$	0.0780	—	0.1267	0.0528	—	0.0918
	(0.1167)	—	(0.1334)	(0.1137)	—	(0.1295)
Δ ln (Establishments)$_{t-1}$	—	−0.0583	−0.1323	—	−0.0550	−0.1323
	—	(0.1904)	(0.2176)	—	(0.1864)	(0.2121)
Precipitation$_t$				yes	yes	yes
Municipality FE	yes	yes	yes	yes	yes	yes
Region-by-year FE	yes	yes	yes	yes	yes	yes
Observations	98,019	98,019	98,019	98,005	98,005	98,005

Notes: Dependent variable is the number of *Heat Days* in year t.
Δ ln (Employment)$_{t-1}$ = ln(Employment$_{t-1}$) − ln(Employment$_{t-2}$) and
Δ ln (Establishments)$_{t-1}$ = ln(Establishments$_{t-1}$) − ln(Establishments$_{t-2}$) denote the municipality one-year lagged log-growth of formal employment and of the establishment stock, respectively. Standard errors are clustered at the municipality level. * p<0.10, ** p<0.05, *** p<0.01.

Table B.3: Reverse Causality: Effect of Economic Activity on $\Delta HeatDays$

	$\Delta HeatDays$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Economic activity									
$\Delta \ln(\text{Employment})_{early}$	0.0518	0.0197	-0.0439	—	—	—	-0.1251	0.0058	0.0364
	(0.1544)	(0.1443)	(0.1446)	—	—	—	(0.1945)	(0.1769)	(0.1785)
$\Delta \ln(\text{Establishments})_{early}$	—	—	—	0.4010	0.0508	-0.2146	0.5440	0.0442	-0.2561
	—	—	—	(0.2870)	(0.2680)	(0.2651)	(0.3625)	(0.3283)	(0.3268)
State FE	yes			yes			yes		
Mesoregion FE		yes			yes			yes	
Microregion FE			yes			yes			yes
Observations	5,420	5,420	5,420	5,420	5,420	5,420	5,420	5,420	5,420

Notes: The dependent variable, $\Delta HeatDays$, is the change in average *Heat Days* between 2000–2009 and 2010–2019 at the municipality level. $\Delta \ln(\text{Employment})_{early}$ and $\Delta \ln(\text{Establishments})_{early}$ denote log-growth in, respectively, formal employment and the establishment stock between the 2000–04 and 2005–09 sub-periods. All specifications control for precipitation and municipality-level baseline (2000) demographic and economic characteristics. Standard errors are clustered at the municipality level. * p<0.10, ** p<0.05, *** p<0.01.

Figure B.1: Effects on establishment size (in %) of 10 *Mild Days*

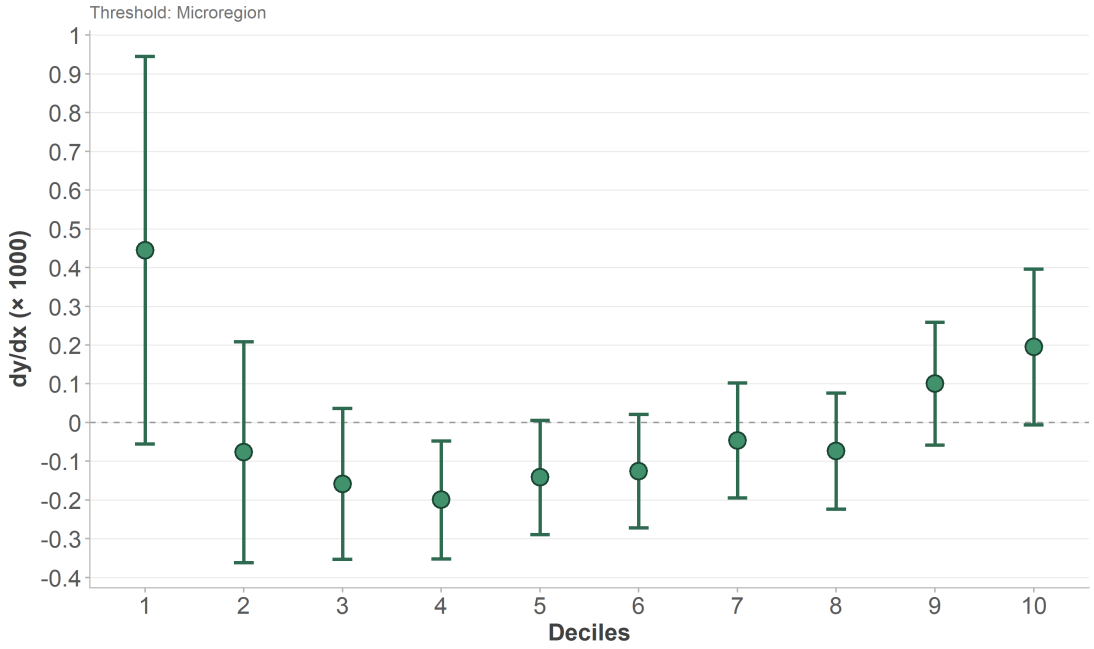
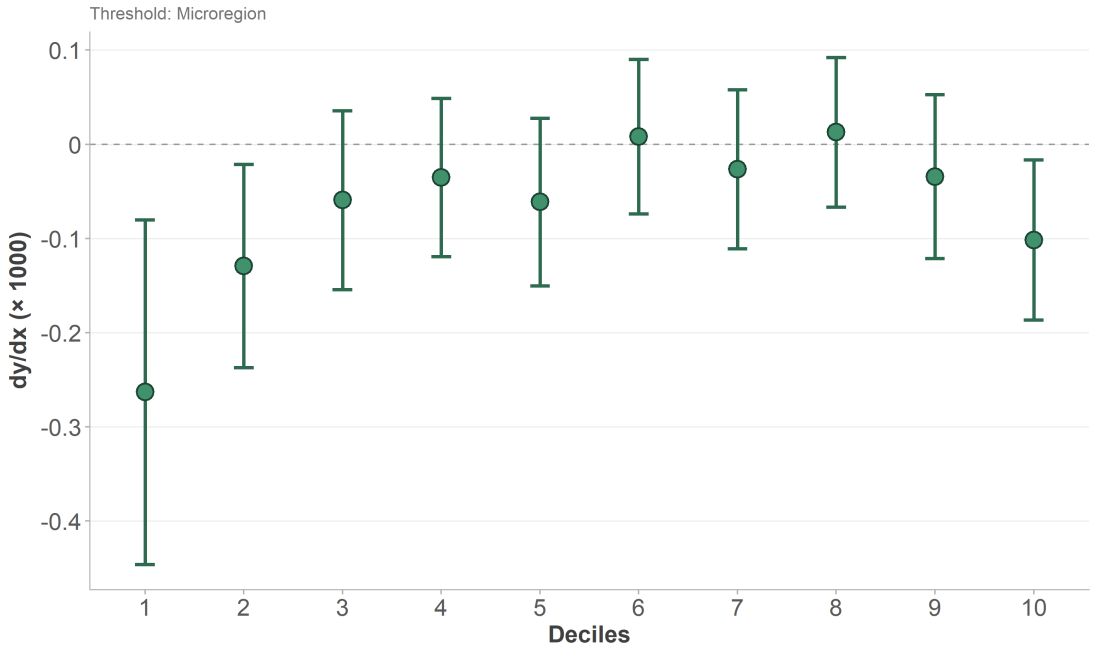


Figure B.2: Effects on establishment exit propensity (in p.p.) of 10 *Mild Days*



Appendix C: AKM and Productivity Proxy Validation

Our main analysis uses residual average wages within local labor markets to proxy establishment productivity. A more precise approach would be to recover establishment fixed effects from AKM regressions (Abowd et al., 1999), which isolate the wage premium paid by establishments from worker composition. This section details our AKM estimation procedure and validates our wage-based measure against the recovered firm effects.

We allow our residual wage proxy for establishment productivity to vary across years. With the goal of keeping a certain level of this variation, we split our 20-year worker-level sample into four non-overlapping five-year periods (2000–2004, 2005–2009, 2010–2014, 2015–2019).[26] We estimate the following AKM-type wage regression for each period separately:

$$ln(wage)_{it} = \alpha_i + \psi_{J(i,t)} + \phi_t + \varepsilon_{it} \tag{4}$$

where $\ln(wage)_{it}$ is the natural logarithm of wages for worker i in year t. As in a standard AKM framework, we include α_i, $\psi_{J(i,t)}$, and ϕ_t, which represent worker, establishment, and year fixed effects, respectively. Following Alvarez et al. (2018), we keep only one observation per worker-year by retaining the highest-paid job among all longest employment spells in any given year.

Identification of establishment fixed effects requires observing workers who move between establishments, restricting estimation to the largest connected set (LCS) – the subset of establishments linked directly or indirectly through worker mobility. Although we were able to link approximately 97–99 percent of all workers in each period to their respective LCS, only 78–84 percent of establishments are encompassed in these sets. Overall, our firm effect is not identified for 17.38% of the full sample of establishments – mainly plants where workers appear only once in the five-year panel.

Table C.1 compares the characteristics of establishments with and without identified firm effects. The excluded establishments are substantially smaller and pay lower wages. Because

[26]The estimation of AKM regressions for sub-periods of the sample allows the fixed effects of workers and establishments that were in multiple largest connected sets to vary across periods, which is a reasonable assumption given that productivity might not be stable over time. However, as previously stated by Lachowska et al. (2023), firm effects are highly temporally persistent.

small establishments are central to our analysis of heterogeneous climate impacts – as they are typically more vulnerable to temperature shocks – excluding them would potentially bias our estimations. This motivates our choice to rely on $\hat{\boldsymbol{\mu}}$ within microregion-industry cells as the primary productivity proxy, which allows us to retain the full universe of formal private establishments.

Table C.1: Characteristics of Establishments by AKM Firm Effect Identification Status

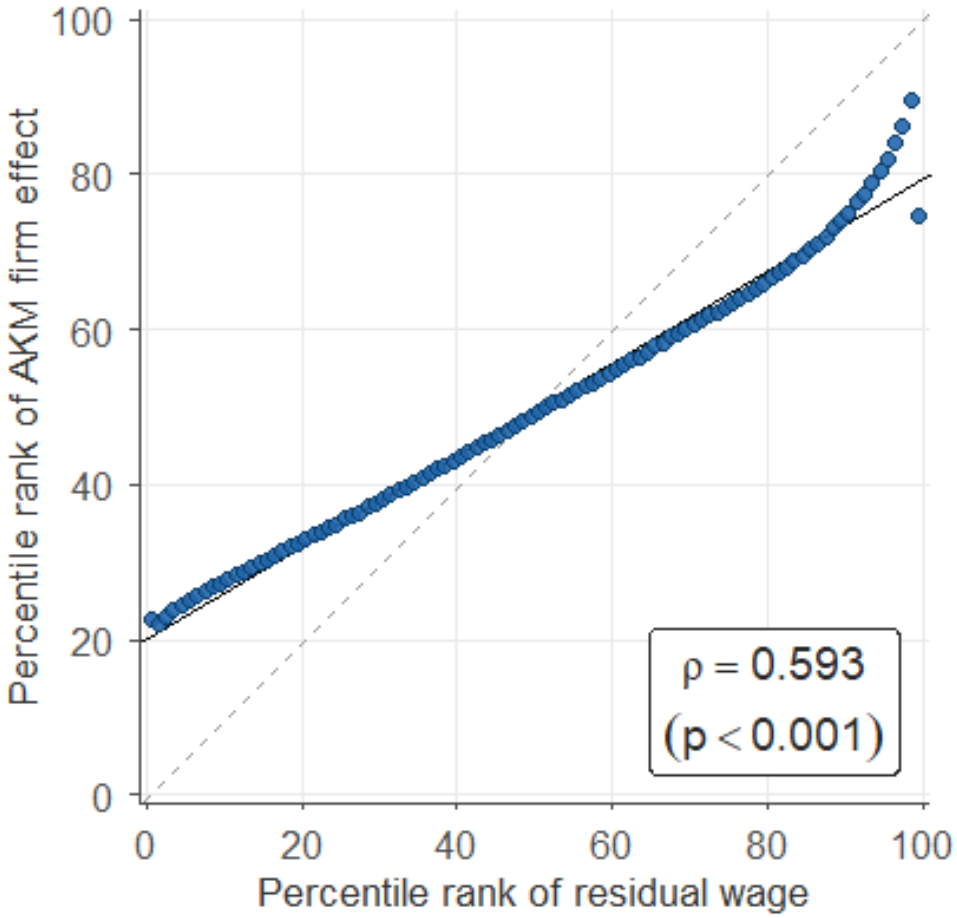
	FE Identified			**FE Not Identified**			**Difference**
Plant Characteristics	*Mean (1)*	*Median*	*SD*	*Mean (2)*	*Median*	*SD*	*(1) − (2)*
Size	13.99	4	87.79	1.61	1	1.83	12.38***
Wage (Monthly BRL)	1579.01	1324.96	1257.79	1336.77	1131.80	1156.58	242.24***
By Sector	*#Plants*	*Share (1)*		*#Plants*	*Share (2)*		*(1) − (2)*
Agriculture	659,899	1.80%	—	221,486	2.87%	—	−1.04 p.p.
Manufacturing	5,834,909	15.91%	—	775,872	10.05%	—	5.86 p.p.
Services	30,179,112	82.29%	—	6,719,837	87.08%	—	−4.79 p.p.
Observations		36,673,920			7,717,195		
Share of the Full Sample		82.62%			17.38%		

Notes: The table compares establishment characteristics between the sample with identified AKM firm effects and the sample where identification was not possible due to lack of worker mobility. The last column reports the difference in means (for the plant characteristics) and in percentage points (for the sectoral composition); significance levels for the difference in means are based on two-sample t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To validate our productivity measure, Figure C.1 plots the percentile rank of an establishment's AKM firm effect against the percentile rank of its residual average wage, for the sample of establishments with identified fixed effects. Both variables are collapsed to the establishment × five-year period level, the unit at which firm effects are estimated. The Spearman correlation between the two measures is a statistically significant 0.593, and the rank–rank relationship is monotonic along the entire distribution. Establishments paying higher residual average wages thus occupy systematically higher positions in the distribu-

tion of wage premia, supporting the use of our proxy as an approximation of establishment productivity.

Figure C.1: Rank–Rank Plot: Establishment Fixed Effect × Residual Average Wage ($\hat{\boldsymbol{\mu}}$)



Notes: Sample restricted to establishments with an identified AKM firm effect, collapsed to the establishment × five-year period level. Each point is the mean percentile rank within one hundredth of the $\hat{\mu}$ distribution; the solid line is the fitted regression over individual observations, whose slope equals the reported Spearman correlation, and the dashed line is the 45-degree line of perfect rank concordance.